

Hospital Management System - Complete Project Specification

Equatorial Guinea Ministry of Health Digital Healthcare Initiative

Project Name: National Healthcare Management System (NHMS) **Client:** Ministry of Health, Social Welfare and Health Infrastructure, Republic of Equatorial Guinea **Strategic Partner:** PERFECT TIMING HOLDING

Project Scope: 5 Major Healthcare Facilities **Project Status:** Ministry-Approved Strategic Initiative

Document Version: 1.0 **Date:** May 2026

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1. Executive Summary

This document outlines the complete specifications for the National Healthcare Management System (NHMS) to be deployed across five major healthcare facilities in Equatorial Guinea. This system is a critical component of the **National Health Development Plan** aimed at ensuring high-quality medical care for the entire population.

Project Objectives:

- Digitize all healthcare operations across government hospitals
- Create unified patient records accessible nationwide
- Improve operational efficiency and reduce waiting times
- Enable data-driven healthcare policy decisions
- Ensure compliance with international healthcare standards
- Facilitate resource optimization across facilities

2. Project Overview

2.1 Vision

To establish a world-class, integrated digital healthcare infrastructure that serves as the foundation for modern medical care delivery in Equatorial Guinea.

2.2 Strategic Importance

- **Government Priority:** Officially recognized as part of National Health Development Plan
- **Ministerial Approval:** Signed and sealed by Minister of Health and Secretary General
- **Population Impact:** Serves the entire population of Equatorial Guinea
- **International Standards:** Aligned with WHO healthcare digitization guidelines
- **Economic Impact:** Foundation for medical tourism and specialized care facilities

2.3 Project Scope

In Scope:

- Complete hospital management software for all 5 facilities
- Cloud-based centralized data infrastructure
- Mobile applications for healthcare workers
- Patient portal for online services
- Ministry dashboard for policy oversight
- Integration with medical equipment and devices
- Training for 500+ healthcare workers
- 24/7 technical support and maintenance

Out of Scope (Phase 2):

- Medical equipment procurement
- Physical infrastructure construction (except software for new facilities)
- Insurance company integrations (future phase)
- International patient referral systems

3. Target Facilities

3.1 General Hospital of Sampaka

- **Location:** Sampaka, Bioko Norte
- **Type:** General Hospital
- **Capacity:** 150+ beds
- **Specialties:** General medicine, surgery, emergency
- **Expected Users:** 80-100 staff members
- **Patient Load:** ~500 patients/day

3.2 Regional Hospital of Malabo

- **Location:** Malabo (Capital City)
- **Type:** Regional Hospital (Largest facility)

- **Capacity:** 250+ beds
- **Specialties:** All major specialties including ICU
- **Expected Users:** 150-200 staff members
- **Patient Load:** ~800 patients/day
- **Special Requirements:** MRI clinic integration, aesthetic surgery unit

3.3 Regional Hospital of Bata

- **Location:** Bata (Mainland)
- **Type:** Regional Hospital + Future University Hospital
- **Capacity:** 200+ beds (expanding to 400+)
- **Specialties:** Teaching hospital with all specialties
- **Expected Users:** 120-150 staff + medical students
- **Patient Load:** ~600 patients/day
- **Special Requirements:** Academic/teaching module, research data management

3.4 Dr. Loeri Comba Polyclinic

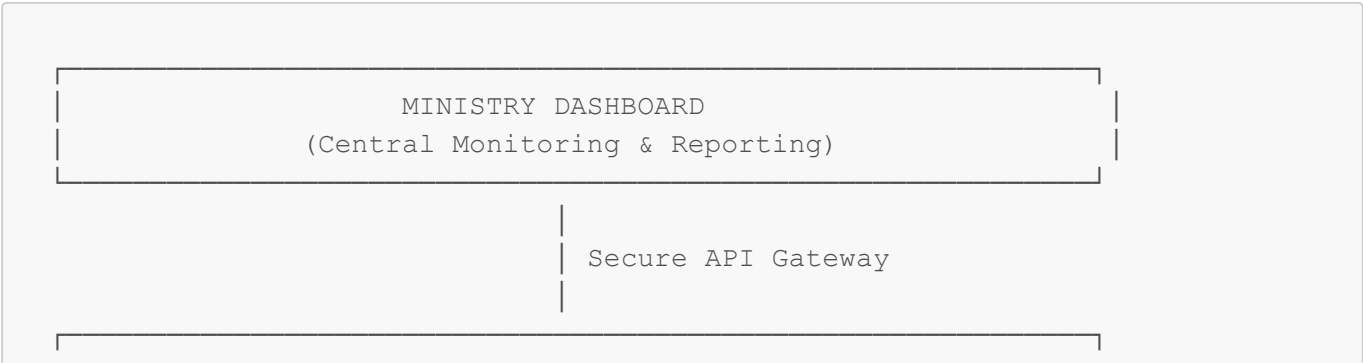
- **Location:** Malabo
- **Type:** Specialized Polyclinic
- **Capacity:** 50 beds
- **Specialties:** Dental clinic, outpatient services, specialized clinics
- **Expected Users:** 40-60 staff members
- **Patient Load:** ~300 patients/day
- **Special Requirements:** Dental management module

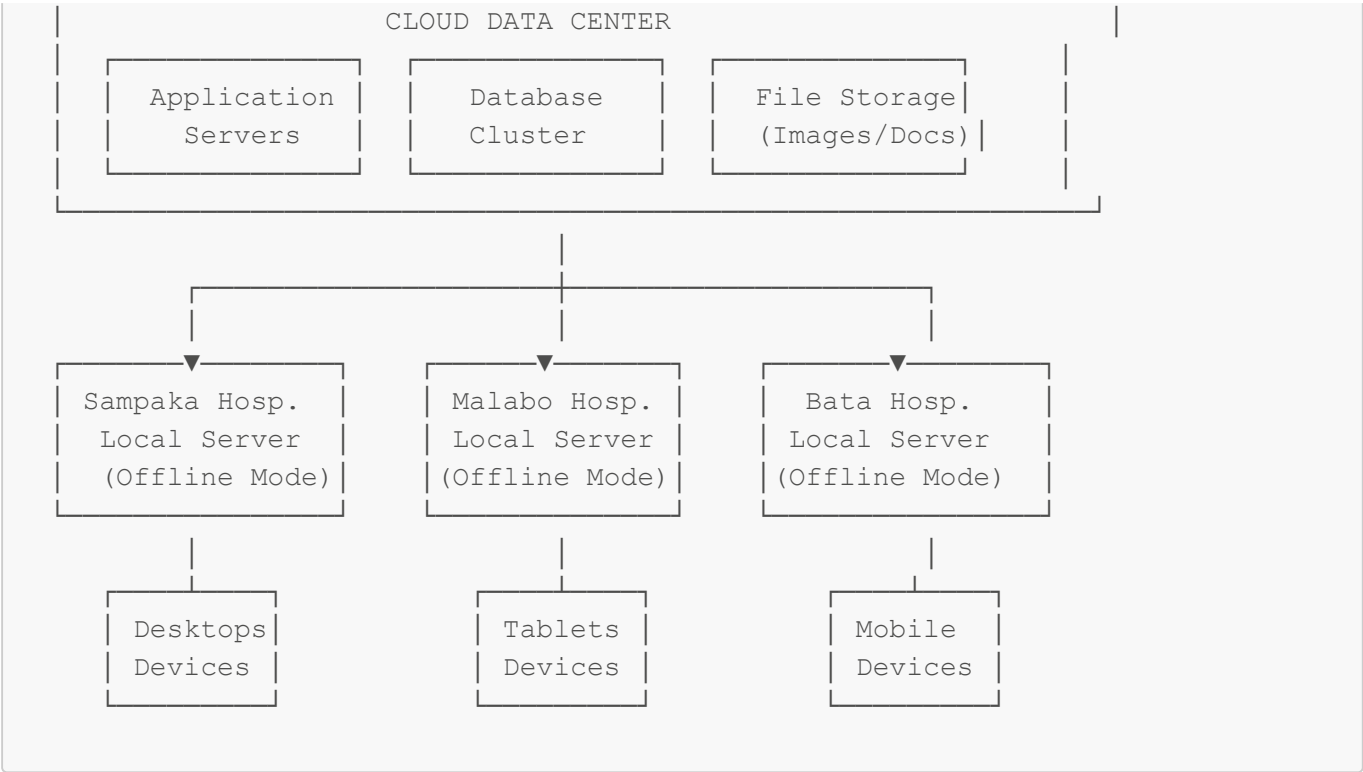
3.5 New INSESO Bata

- **Location:** Bata
- **Type:** Social Security Healthcare Facility
- **Capacity:** 100+ beds
- **Specialties:** Social security beneficiary care
- **Expected Users:** 60-80 staff members
- **Patient Load:** ~400 patients/day
- **Special Requirements:** Social security integration, benefit verification

4. System Architecture

4.1 Architecture Overview





4.2 Technology Stack

Frontend:

- **Web Application:** React.js / Vue.js (Progressive Web App)
- **Mobile Apps:** React Native (iOS & Android)
- **Admin Dashboard:** Angular / Next.js
- **Responsive Design:** Bootstrap 5 / Tailwind CSS

Backend:

- **API Server:** Node.js (Express) / Python (FastAPI)
- **Authentication:** JWT + OAuth 2.0
- **API Documentation:** OpenAPI/Swagger
- **Microservices:** Docker containerization

Database:

- **Primary Database:** PostgreSQL (patient records, transactions)
- **Cache Layer:** Redis (session management, performance)
- **Document Storage:** MongoDB (unstructured data, logs)
- **Data Warehouse:** Apache Spark (analytics, reporting)

File Storage:

- **Images/Documents:** AWS S3 / MinIO (self-hosted)
- **Medical Images:** DICOM-compliant PACS system
- **Backup Storage:** Distributed backup across facilities

Infrastructure:

- **Cloud Provider:** AWS / Microsoft Azure / Private Cloud

- **CDN:** CloudFlare (for assets)
- **Load Balancer:** NGINX / HAProxy
- **Monitoring:** Prometheus + Grafana
- **Logging:** ELK Stack (Elasticsearch, Logstash, Kibana)

4.3 Network Architecture

Connectivity:

- Primary: Fiber optic connections to each facility
- Backup: 4G/5G cellular backup
- Offline Mode: Local servers with auto-sync when online
- VPN: Secure tunnels between facilities and central data center

Bandwidth Requirements:

- Malabo Regional: 100 Mbps minimum
- Bata Regional: 100 Mbps minimum
- Other facilities: 50 Mbps minimum

5. Complete Module Specifications

MODULE 1: Patient Management System (PMS)

5.1.1 Patient Registration

Functionality:

- National Patient ID generation (unique across all hospitals)
- Biometric capture (photo, fingerprint optional)
- Demographic data collection
- Next of kin / emergency contact
- Insurance/payment method registration
- Consent form management

Data Fields:

Personal Information:

- National ID Number

- Full Name (First, Middle, Last)

- Date of Birth

- Gender

- Blood Type

- Nationality

- Preferred Language (Spanish/French/Portuguese/Fang)

- Photo

Contact Information:

- Phone Number (Primary/Secondary)

- Email Address

```
- Physical Address
- GPS Coordinates (for rural areas)

Emergency Contact:
- Name, Relationship, Phone

Insurance/Payment:
- Insurance Provider
- Policy Number
- Payment Method (Cash/Insurance/Government/Social Security)
- Employer (if applicable)

Medical Information:
- Known Allergies
- Chronic Conditions
- Current Medications
- Blood Type
- Organ Donor Status
```

Features:

- Duplicate patient detection (fuzzy matching)
- Merge patient records
- Patient search (ID, name, phone, partial matching)
- QR code generation for patient ID card
- Export patient demographics

User Stories:

```
US-PM-001: As a receptionist, I want to register a new patient in under 5
minutes
US-PM-002: As a receptionist, I want to search for existing patients by
name or ID
US-PM-003: As a doctor, I want to see patient allergies prominently
displayed
US-PM-004: As a patient, I want to receive a digital ID card on my phone
```

5.1.2 Electronic Medical Records (EMR)**Functionality:**

- Complete medical history timeline
- Problem list (active/resolved diagnoses)
- Medication history
- Immunization records
- Surgical history
- Family medical history
- Social history (smoking, alcohol, occupation)
- Clinical notes (SOAP format)

- Attachment management (scanned documents)

Clinical Documentation:

SOAP Notes Structure:

- Subjective: Chief complaint, patient narrative
- Objective: Vital signs, examination findings
- Assessment: Diagnosis, differential diagnosis
- Plan: Treatment plan, prescriptions, follow-up

Templates for Common Conditions:

- Hypertension follow-up
- Diabetes management
- Prenatal care
- Pediatric well-child visit
- Emergency trauma

Features:

- Version control for all records
- Audit trail (who accessed/modified when)
- Digital signature for doctors
- Voice-to-text note entry
- Template-based documentation
- ICD-10 diagnosis coding
- Clinical decision support alerts

Alerts System:

- Drug allergies
- Drug-drug interactions
- Abnormal vital signs
- Overdue vaccinations
- Missing critical information

User Stories:

US-EMR-001: As a doctor, I want to see a patient's complete history in one screen

US-EMR-002: As a doctor, I want to use templates to speed up documentation

US-EMR-003: As a nurse, I want to record vital signs that automatically alert if abnormal

US-EMR-004: As an auditor, I want to see who accessed sensitive patient records

5.1.3 Vital Signs Monitoring

Captured Vitals:

- Blood Pressure (systolic/diastolic)
- Heart Rate
- Respiratory Rate
- Temperature (Celsius/Fahrenheit)
- Oxygen Saturation (SpO2)
- Blood Glucose
- Weight / Height / BMI
- Pain Scale (0-10)
- Consciousness Level (Glasgow Coma Scale)

Features:

- Automatic device integration (blood pressure monitors, thermometers)
- Manual entry with validation
- Graphical trending (last 7 days, 30 days, all time)
- Automatic alerts for abnormal values
- Pediatric vs adult normal ranges
- Export to PDF

Alert Thresholds (Configurable):

```
Critical Alerts:  
- Blood Pressure: >180/120 or <90/60  
- Heart Rate: >120 or <50  
- SpO2: <90%  
- Temperature: >39.5°C or <35°C  
- Blood Glucose: >300 or <70 mg/dL
```

MODULE 2: Appointment Scheduling System

5.2.1 Appointment Management

Functionality:

- Multi-channel booking (in-person, phone, online, mobile app)
- Doctor availability calendar
- Department/specialty-based scheduling
- Time slot management (configurable duration)
- Recurring appointment series
- Group appointments (e.g., prenatal classes)
- Walk-in queue management

Scheduling Rules:

- Slot duration: 15/30/45/60 minutes (configurable per doctor)
- Overbooking allowance (emergency slots)
- Break time blocking

- Holiday/leave management
- Multi-location support (doctor works at multiple facilities)

Features:

- Drag-and-drop calendar interface
- Color-coded appointment types
- Patient reminder system (SMS/Email)
- Automatic waitlist management
- No-show tracking
- Cancellation management
- Rescheduling with conflict detection

Appointment Types:

- New patient consultation
- Follow-up visit
- Procedure/surgery
- Lab test appointment
- Imaging appointment
- Vaccination
- Health screening
- Telemedicine consultation

Notification Triggers:**Appointment Reminders:**

- 7 days before (for surgeries)
- 24 hours before
- 2 hours before (SMS)

Notifications to Staff:

- New appointment booked
- Patient checked in
- Patient is waiting (>15 min past appointment time)
- Appointment cancelled

User Stories:

US-AS-001: As a patient, I want to book an appointment online without calling

US-AS-002: As a receptionist, I want to see which doctors are available today

US-AS-003: As a doctor, I want to block time for administrative work

US-AS-004: As a patient, I want to receive SMS reminder before my appointment

US-AS-005: As a manager, I want to see no-show rates per doctor/department

5.2.2 Queue Management

Functionality:

- Digital check-in (QR code scan or ID number)
- Real-time queue display boards
- Priority queue (emergency, pregnant women, elderly, disabled)
- Estimated wait time calculation
- Queue transfer between doctors
- Virtual queue (patients can wait elsewhere)

Features:

- TV display for waiting areas
- Audio announcement system integration
- Mobile app queue status
- Average wait time analytics
- Queue abandonment tracking

MODULE 3: Doctor/Staff Management

5.3.1 Staff Directory

Staff Information:

Personal Details:

- Employee ID

- Full Name

- Photo

- Date of Birth

- Contact Information

Professional Details:

- Job Title/Position

- Department

- Specialty

- License Number

- License Expiry Date

- Qualifications/Degrees

- Languages Spoken

- Years of Experience

Employment Details:

- Employment Type (Full-time/Part-time/Contract)

- Date of Joining

- Contract End Date

- Salary Grade

- Reporting Manager

- Work Locations

Credentials:

- Medical License (upload scan)
- Certifications
- Training Records
- Performance Reviews

Features:

- Staff search and filter
- Org chart visualization
- License expiry alerts
- Birthday/anniversary reminders
- Performance tracking
- Competency management

5.3.2 Shift Scheduling & Attendance**Functionality:**

- Shift pattern creation (day/night/rotating)
- Automatic shift roster generation
- Shift swap requests
- On-call duty scheduling
- Leave management (annual, sick, maternity)
- Time and attendance tracking
- Overtime calculation

Leave Types:

- Annual Leave
- Sick Leave
- Maternity/Paternity Leave
- Study Leave
- Unpaid Leave
- Compassionate Leave

Features:

- Calendar view (daily/weekly/monthly)
- Conflict detection (understaffing alerts)
- Mobile shift viewing
- Shift handover notes
- Coverage ratio monitoring (minimum staff per department)

Attendance Tracking:

- Biometric clock-in/out
- Manual attendance entry
- Late arrival alerts

- Absence management
- Attendance reports (monthly/yearly)

User Stories:

```
US-SM-001: As a nurse, I want to request shift swap with a colleague
US-SM-002: As a manager, I want to see who is on duty right now
US-SM-003: As HR, I want to track leave balances for all staff
US-SM-004: As a doctor, I want to see my schedule on my mobile phone
```

5.3.3 Performance Management

KPIs Tracked:

- Patient satisfaction scores
- Average consultation time
- Number of patients seen per day
- Prescription accuracy
- Complication rates
- Attendance and punctuality
- Continuing education hours

Features:

- Performance dashboard per staff member
- Goal setting and tracking
- 360-degree feedback
- Training needs identification
- Certification tracking

MODULE 4: Pharmacy Management System

5.4.1 Medication Inventory

Inventory Data:

```
Medication Information:
- Drug Name (Generic + Brand)
- Drug Classification
- Manufacturer
- Strength/Dosage
- Form (Tablet/Capsule/Syrup/Injection)
- NDC/Barcode
- Therapeutic Class
- Storage Requirements
- Prescription vs OTC
```

Stock Information:

- Current Stock Level
- Reorder Level
- Maximum Stock Level
- Storage Location (Shelf/Bin)
- Batch Number
- Manufacturing Date
- Expiry Date
- Purchase Price
- Selling Price
- Supplier Details

Features:

- Real-time stock levels
- Expiry date tracking (alerts 6 months before)
- Batch tracking (FIFO/FEFO)
- Barcode scanning
- Stock transfer between hospitals
- Dead stock identification
- ABC/VEN analysis
- Temperature monitoring (refrigerated drugs)

Alerts:

- Low stock warning (below reorder level)
- Expiry alert (6 months, 3 months, 1 month, expired)
- Overstock warning
- Out of stock alert
- Temperature deviation (for refrigerated items)

5.4.2 Prescription Management**e-Prescription Features:**

- Drug search (generic/brand name)
- Dosage calculator (based on weight/age)
- Favorite prescriptions (save common combinations)
- Drug interaction checker
- Allergy checker
- Duplicate therapy alert
- Formulary compliance
- Electronic signature

Prescription Format:**Header:**

- Patient Name, Age, Gender, Weight
- Patient ID, Allergies (prominently displayed)

- Doctor Name, License Number
- Hospital Name, Date

Medications (per line):

- Drug Name
- Strength
- Form
- Quantity
- Dosage Instructions (e.g., "1 tablet twice daily after meals")
- Duration (e.g., "for 7 days")
- Special Instructions

Footer:

- Digital signature
- QR code (for verification)
- Prescription number

Dispensing Workflow:

1. Pharmacist receives e-prescription
2. Verify patient identity
3. Check drug interactions (system alerts)
4. Dispense medication (scan barcode)
5. System deducts from inventory
6. Print medication label
7. Patient counseling (document in system)
8. Patient signature (received medications)
9. Generate bill automatically

Features:

- Prescription queue (pending/in-progress/completed)
- Partial dispensing (if full quantity unavailable)
- Substitution management (generic for brand)
- Medication counseling checklist
- Prescription history
- Refill management
- Controlled substance tracking

User Stories:

US-PM-001: As a doctor, I want to prescribe medications with auto-completion

US-PM-002: As a doctor, I want to see alerts if prescribing a drug patient is allergic to

US-PM-003: As a pharmacist, I want to see all pending prescriptions in a queue

US-PM-004: As a pharmacist, I want to document patient counseling

```
US-PM-005: As a patient, I want to receive medication instructions in my language
```

5.4.3 Procurement & Purchase Orders

Functionality:

- Automatic reorder point triggering
- Purchase requisition workflow
- Supplier management
- Purchase order generation
- Goods receipt management
- Invoice matching (3-way match)
- Payment tracking
- Supplier performance tracking

Purchase Order Workflow:

1. System generates alert: Drug X below reorder level
2. Pharmacist reviews and creates requisition
3. Pharmacy manager approves requisition
4. Procurement officer selects supplier (best price)
5. System generates purchase order
6. PO sent to supplier (email/portal)
7. Goods received, scanned into system
8. Invoice received and matched
9. Payment processed
10. Stock updated automatically

Supplier Management:

- Supplier database
- Lead time tracking
- Price comparison
- Quality ratings
- Payment terms
- Contact information
- Contract management

5.4.4 Blockchain-Based Drug Traceability (Optional Module)

Overview: Optional blockchain-powered pharmaceutical supply chain traceability system to verify drug authenticity, origin, and prevent counterfeit medications from entering the healthcare system. This module provides an immutable, transparent record of every drug's journey from manufacturer to patient.

Strategic Benefits:

- **Combat Counterfeit Drugs:** WHO estimates 10% of drugs in developing countries are counterfeit - blockchain verification prevents fake drugs
- **Regulatory Compliance:** Meets international pharmaceutical traceability standards (FDA DSCSA, EU FMD)
- **Patient Safety:** Ensures patients receive genuine, properly stored medications
- **Recall Management:** Instantly identify and trace affected batches during drug recalls
- **Supply Chain Transparency:** Full visibility into drug journey across borders and organizations
- **Audit Trail:** Immutable record for regulatory inspections and quality assurance

How It Works:

Drug Journey on Blockchain:

1. MANUFACTURER (Pfizer, Spain)
 - ↓ Records on blockchain:
 - Drug: Amoxicillin 500mg
 - Batch Number: PFZ-AMX-2026-05-12345
 - Manufacturing Date: 15 May 2026
 - Expiry Date: 15 May 2028
 - Quantity: 10,000 tablets
 - Quality Certificates: [Attached]
 - Blockchain Hash: 0x3f7a...b2c4
2. INTERNATIONAL DISTRIBUTOR (MedSupply Global)
 - ↓ Scans QR code, records on blockchain:
 - Received: 20 May 2026
 - Storage: Temperature controlled (15-25°C)
 - Transit Time: 5 days
 - Customs Clearance: Verified
 - Blockchain Hash: 0x8d2e...f9a1
3. LOCAL DISTRIBUTOR (Equatorial Guinea MedCare)
 - ↓ Scans QR code, records on blockchain:
 - Received: 25 May 2026
 - Location: Malabo Warehouse
 - Storage Conditions: Monitored
 - Inspector: Ministry of Health approved
 - Blockchain Hash: 0x1c5b...3e7d
4. HOSPITAL PHARMACY (Regional Hospital Malabo)
 - ↓ Scans QR code, records on blockchain:
 - Received: 28 May 2026
 - Pharmacist: Maria Obiang
 - Storage Location: Shelf A-12
 - Temperature: 22°C (within range)
 - Blockchain Hash: 0x9a4f...d8c2
5. DISPENSING TO PATIENT
 - ↓ Pharmacist scans before dispensing:
 - ✓ VERIFICATION PASSED

Drug: Amoxicillin 500mg
Manufacturer: Pfizer (Spain) ✓ VERIFIED
Batch: PFZ-AMX-2026-05-12345
Expiry: 15 May 2028 (Valid for 2 years)

Journey:
Pfizer → MedSupply Global → EG MedCare → Malabo Hospital

Transit Time: 13 days
Storage Conditions: ✓ MAINTAINED
Temperature Deviations: None

Authenticity: ✓ GENUINE
Counterfeit Risk: 0%

Safe to Dispense: YES ✓

Patient: Juan Nguema Mbomio
Prescribed by: Dr. Milagrosa Ondo
Date: 30 May 2026
Blockchain Hash: 0x7e2a...f1b9

If Counterfeit Drug Detected:

⚠ COUNTERFEIT ALERT

Drug: Amoxicillin 500mg (SUSPICIOUS)
Batch: ABC-123-FAKE

✗ VERIFICATION FAILED

Issues Detected:

- Batch number not found in manufacturer records
- QR code does not match blockchain registry
- No temperature monitoring data
- Origin cannot be verified
- Missing quality certificates

Risk Level: HIGH
Authenticity: ✗ COUNTERFEIT SUSPECTED

⚠ DO NOT DISPENSE

⚠ QUARANTINE IMMEDIATELY

⚠ REPORT TO MINISTRY OF HEALTH

Incident ID: INC-2026-0542
Auto-reported to: National Drug Authority

Core Features:

1. QR Code Generation & Scanning

- Each drug package has unique QR code
- Contains encrypted blockchain transaction ID
- Scannable by smartphone or handheld scanner
- Instant verification in < 2 seconds

2. Immutable Blockchain Ledger

- Every transaction permanently recorded
- Cannot be altered or deleted
- Distributed ledger (no single point of failure)
- Cryptographically secured
- Timestamp verified

3. Real-Time Verification

- Scan drug package at any point
- Instant verification: genuine vs counterfeit
- See complete journey (manufacturer to current location)
- Check storage conditions (temperature, humidity)
- Verify expiry date

4. Temperature & Storage Monitoring

- IoT sensors track storage conditions
- Automatic blockchain recording
- Alerts if temperature exceeds safe range
- Cold chain compliance for refrigerated drugs

5. Batch Recall Management

- Identify all locations of recalled batch instantly
- Auto-alert hospitals with affected stock
- Track which patients received recalled drugs
- Complete audit trail for regulatory reporting

6. Manufacturer Integration

- Major pharmaceutical companies integrate at source
- API for automatic blockchain registration
- Bulk upload for large shipments
- Digital certificates attached

7. Customs & Border Control Integration

- Verify drugs at border entry
- Auto-flag suspicious shipments
- Prevent counterfeit imports

- Regulatory compliance documentation

8. Patient-Level Tracking

- Record which patient received which batch
- Pharmacogenomics data (future)
- Adverse reaction correlation
- Personalized medicine support

9. Analytics Dashboard

National Drug Traceability Dashboard

Drugs Tracked (This Month): 1,234,567 units

Scans Performed: 4,567,890

Counterfeit Detected: 47 instances (0.004%)

Counterfeit Value Prevented: \$245,000

Top Counterfeited Drugs:

1. Antibiotics (Amoxicillin, Azithromycin)

2. Antimalarials (Artemisinin-based)

3. Pain relievers (Tramadol)

Supply Chain Efficiency:

Average Transit Time: 18 days

Temperature Violations: 12 incidents (0.001%)

Expired Drug Prevention: 234 units caught

Blockchain Network:

Total Transactions: 15,234,890

Network Nodes: 25 (hospitals, distributors, manufacturers)

Uptime: 99.98%

Average Verification Time: 1.2 seconds

10. Ministry of Health Oversight

- Real-time visibility into entire drug supply chain
- Identify supply bottlenecks
- Detect counterfeit patterns
- Policy decisions based on data
- Regulatory enforcement

Technical Implementation:

Blockchain Platform Options:

Option 1: Hyperledger Fabric (Recommended)

- Private/permissioned blockchain

- Fast transaction speed (1000+ TPS)

- Low cost (no cryptocurrency required)
- Enterprise-grade security
- Used by: Walmart, IBM Food Trust

Option 2: Ethereum (Public Blockchain)

- Public, transparent ledger
- Smart contracts for automation
- Higher transaction costs (gas fees)
- Global accessibility

Option 3: VeChain (Healthcare-Specific)

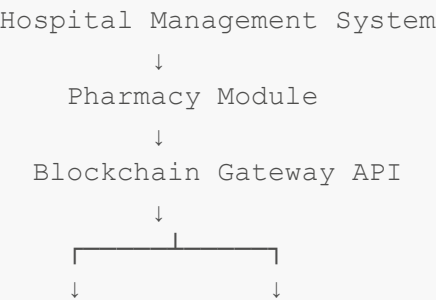
- Designed for supply chain
- IoT sensor integration
- Lower costs than Ethereum
- Healthcare focus

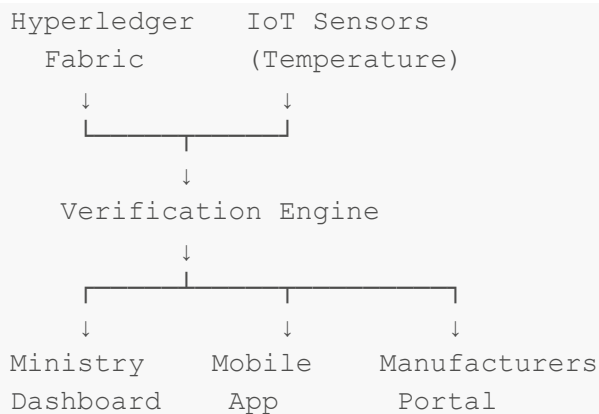
Data Stored on Blockchain (Per Drug Package):

```
{
  "drugId": "PFZ-AMX-2026-05-12345-0001",
  "drugName": "Amoxicillin",
  "strength": "500mg",
  "form": "Tablet",
  "manufacturer": {
    "name": "Pfizer S.A.",
    "location": "Madrid, Spain",
    "license": "ES-PHARMA-001234",
    "gmpCertified": true
  },
  "batchNumber": "PFZ-AMX-2026-05-12345",
  "manufacturingDate": "2026-05-15",
  "expiryDate": "2028-05-15",
  "quantity": 100,
  "qrCode": "https://verify.nhms.gq/PFZ-AMX-2026-05-12345-0001",
  "journey": [
    {
      "timestamp": "2026-05-15T10:30:00Z",
      "location": "Pfizer Factory, Madrid",
      "actor": "Pfizer Manufacturing",
      "action": "MANUFACTURED",
      "temperature": "20°C",
      "hash":
"0x3f7a2e8b9c4d5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a"
    },
    {
      "timestamp": "2026-05-20T14:20:00Z",
      "location": "MedSupply Warehouse, Barcelona",
      "actor": "MedSupply Global",
      "action": "RECEIVED",
      "temperature": "18°C",
      "hash":
"0x8d2e1f3a4b5c6d7e8f9a0b1c2d3e4f5a6b7c8d9e0f1a2b3c4d5e6f7a8b9c0d1e"
```

```
    },
    {
      "timestamp": "2026-05-25T09:15:00Z",
      "location": "Malabo Port, Equatorial Guinea",
      "actor": "EG MedCare Distributors",
      "action": "CUSTOMS_CLEARED",
      "temperature": "22°C",
      "customsOfficer": "Inspector Rodriguez",
      "hash":
"0x1c5b3e7d9f0a2b4c6d8e0f1a3b5c7d9e1f3a5b7c9d1e3f5a7b9c1d3e5f7a9b1c"
    },
    {
      "timestamp": "2026-05-28T11:00:00Z",
      "location": "Regional Hospital Malabo - Pharmacy",
      "actor": "Pharmacist Maria Obiang",
      "action": "RECEIVED_AT_HOSPITAL",
      "temperature": "22°C",
      "storageLocation": "Shelf A-12",
      "hash":
"0x9a4f6d8e0c2a4b6c8d0e2f4a6b8c0d2e4f6a8b0c2d4e6f8a0b2c4d6e8f0a2b4c"
    },
    {
      "timestamp": "2026-05-30T15:45:00Z",
      "location": "Regional Hospital Malabo - Pharmacy",
      "actor": "Pharmacist Maria Obiang",
      "action": "DISPENSED",
      "patientId": "2026-MLO-00542",
      "prescribedBy": "Dr. Milagrosa Ondo",
      "quantity": 21,
      "hash":
"0x7e2a5b3c9d1e4f6a8b0c2d4e6f8a0b2c4d6e8f0a2b4c6d8e0f2a4b6c8d0e2f4a"
    }
  ],
  "qualityCertificates": [
    "ISO-9001-2026",
    "GMP-EU-2026",
    "WHO-Prequalified"
  ],
  "verificationStatus": "VERIFIED",
  "counterfeitsDetected": 0
}
```

System Integration:





User Workflow:

For Pharmacist:

1. Receive drug shipment
2. Open mobile app or use handheld scanner
3. Scan QR code on drug package
4. System displays verification result (2 seconds)
5. If verified: Accept into inventory
6. If suspicious: Quarantine & report
7. System auto-updates blockchain

For Doctor/Pharmacist (Dispensing):

1. Prescription created in system
2. Pharmacist retrieves drug from shelf
3. Scans QR code before dispensing
4. System verifies:
 - Authenticity ✓
 - Not expired ✓
 - Storage conditions maintained ✓
 - Not recalled ✓
5. If all OK: Dispense to patient
6. System records patient received this specific batch

For Patient (Optional Patient App):

1. Patient receives medication
2. Opens patient app, scans QR code
3. Sees:
 - Drug name and dosage
 - Manufacturer information
 - Expiry date
 - Complete journey

- Instructions in their language
- Video on how to take medication

Cost-Benefit Analysis:

Costs:

- Blockchain platform setup: \$25,000 (one-time)
- Integration with pharmacy module: \$15,000
- QR code printing system: \$5,000
- Annual blockchain transaction fees: \$5,000/year
- Training: Included in main training **Total: \$50,000 (one-time) + \$5,000/year**

Benefits:

- Prevent counterfeit drugs: ~\$500,000/year (estimated)
- Reduce medication errors: ~\$200,000/year
- Improve patient trust: Priceless
- Regulatory compliance: Avoids fines
- Enhanced reputation: Attracts international partnerships

ROI: 14x in first year

Regulatory Alignment:

- WHO Model Quality Assurance System
- FDA Drug Supply Chain Security Act (DSCSA)
- EU Falsified Medicines Directive (FMD)
- Equatorial Guinea National Drug Authority standards

Pilot Implementation:

Phase 1 (Month 6-9):

- Implement at 1 hospital (Malabo Regional)
- Track 10 high-risk drugs (antibiotics, antimalarials)
- Partner with 2-3 major manufacturers
- Train 20 pharmacy staff

Phase 2 (Month 10-12):

- Expand to all 5 hospitals
- Track all controlled substances
- Add temperature monitoring
- Ministry dashboard operational

Phase 3 (Month 13-18):

- National rollout
- All drugs tracked
- Manufacturer integration (Pfizer, Novartis, etc.)
- Patient app with drug verification

Success Metrics:

- Counterfeit detection rate
- % of drugs tracked (target: 80%+ by end of Year 1)
- Verification scan time (target: < 2 seconds)
- Counterfeit monetary value prevented
- Recall response time (target: < 24 hours)
- Patient trust score

Mobile App Features:

- Scan QR code with phone camera
- Instant verification (green ✓ or red ✗)
- View drug journey on map
- Report suspicious drugs
- Receive alerts for recalls
- Multilingual support

Future Enhancements:

- AI-powered counterfeit pattern detection
- Integration with customs/border control
- Cross-border drug tracking (CEMAC region)
- Smart contracts for automatic payments
- Predictive analytics for supply shortages
- Patient adherence tracking

Strategic Positioning:

This positions Equatorial Guinea as a **regional leader in pharmaceutical safety** and could become a model for other African nations. The Ministry could potentially offer this as a service to neighboring countries through CEMAC (Central African Economic and Monetary Community).

Partnership Opportunities:

- WHO collaboration on drug safety
- Pharmaceutical manufacturers (Pfizer, Novartis) partnership
- International donors (USAID, Global Fund)
- Technology providers (IBM, VeChain, Hyperledger)

MODULE 5: Laboratory Information System (LIS)

5.5.1 Test Order Management

Functionality:

- Test catalog (all available tests)
- Test ordering by doctors
- Test grouping (panels: CBC, LFT, RFT, etc.)
- Urgent/Routine priority

- Specimen requirements
- Patient preparation instructions
- Cost display
- Insurance coverage check

Common Test Panels:

Hematology:

- Complete Blood Count (CBC)
- ESR
- Blood Film
- Coagulation Studies (PT, PTT, INR)

Clinical Chemistry:

- Liver Function Tests (LFT)
- Renal Function Tests (RFT)
- Lipid Profile
- Blood Glucose (Fasting/Random/HbA1c)
- Electrolytes
- Thyroid Function Tests

Microbiology:

- Blood Culture
- Urine Culture
- Stool Culture
- Sensitivity Testing

Serology:

- HIV Test
- Hepatitis Markers
- Malaria Test
- Pregnancy Test

Pathology:

- Biopsy
- Cytology
- Histopathology

Test Order Details:

- Patient Information
- Requesting Doctor
- Test(s) Ordered
- Clinical Notes/Indication
- Priority (Routine/Urgent/STAT)
- Specimen Type
- Collection Date/Time
- Fasting Status
- Special Instructions

5.5.2 Specimen Management

Specimen Tracking:

- Barcode generation at collection
- Specimen labeling (name, ID, date, time, test)
- Chain of custody tracking
- Specimen status (collected, received, processed, discarded)
- Storage location tracking
- Temperature monitoring

Specimen Types:

- Blood (whole, serum, plasma)
- Urine
- Stool
- Sputum
- CSF (Cerebrospinal Fluid)
- Tissue samples
- Swabs

Workflow:

1. Doctor orders test
2. Nurse collects specimen
3. Barcode label printed and applied
4. Specimen sent to lab
5. Lab receives and scans barcode
6. Status: "Received in lab"
7. Technician processes specimen
8. Status: "In progress"
9. Results entered
10. Status: "Completed"
11. Results reviewed by pathologist
12. Results approved and released
13. Doctor notified

Features:

- Reject specimen (insufficient quantity, hemolyzed, contaminated)
- Re-collection request
- Specimen tracking across departments
- Time-sensitive specimen alerts

5.5.3 Result Entry & Reporting

Result Entry Methods:

- Manual entry

- Interface with lab equipment (auto-import)
- Bulk import (CSV/Excel)
- Voice-to-text entry

Result Format:

Test Report Header:

- Patient Demographics
- Specimen Information
- Test Name

Results Table:

Test Parameter	Result	Reference Range	Flag
Hemoglobin	12.5	12.0-16.0 g/dL	
WBC Count	15.2	4.0-11.0 K/uL	HIGH ↑
Platelets	180	150-400 K/uL	

Interpretation/Comments:

- Free text field for pathologist remarks

Footer:

- Tested by: [Technician Name]
- Reviewed by: [Pathologist Name]
- Date & Time
- Digital Signatures

Features:

- Auto-flagging (High/Low/Critical)
- Delta checking (compare with previous results)
- Reference ranges (age/gender-specific)
- Graphical trending
- Critical value notification (immediate SMS/call to doctor)
- Result validation workflow
- Addendum/correction functionality
- Report templates

Critical Values (Auto-Alert):

- Hemoglobin: <7 or >20 g/dL
- WBC: <2 or >30 K/uL
- Platelets: <50 K/uL
- Glucose: <50 or >500 mg/dL
- Potassium: <2.5 or >6.5 mEq/L

- Creatinine: >5 mg/dL
- Positive Blood Culture

User Stories:

US-LIS-001: As a doctor, I want to order multiple tests at once (panel)
US-LIS-002: As a nurse, I want to print specimen labels with barcodes
US-LIS-003: As a lab tech, I want results to auto-import from the analyzer
US-LIS-004: As a pathologist, I want to review all abnormal results before release
US-LIS-005: As a doctor, I want to receive instant alerts for critical results

5.5.4 Quality Control

Functionality:

- Daily QC testing (control samples)
- Equipment calibration tracking
- Reagent lot tracking
- Proficiency testing management
- Error/incident reporting
- Turnaround time monitoring

QC Tracking:

- Control levels (Low/Normal/High)
- Levy-Jennings charts
- Westgard rules violations
- Equipment maintenance logs

MODULE 6: Radiology Information System (RIS) & PACS

5.6.1 Imaging Order Management

Imaging Modalities:

- X-Ray (Plain Radiography)
- Ultrasound
- CT Scan
- MRI
- Mammography
- Fluoroscopy
- Nuclear Medicine (if available)

Order Workflow:

1. Doctor orders imaging study
2. Clinical indication required
3. Previous imaging history shown
4. Radiation dose tracking (cumulative)
5. Pregnancy screening (for females)
6. Contrast allergy check
7. Renal function check (for contrast studies)
8. Patient preparation instructions
9. Appointment scheduled
10. Consent form signed (for procedures)

Order Details:

- Patient Information
- Requesting Physician
- Clinical Indication
- Body Part/Region
- Imaging Protocol
- Contrast Required? (Yes/No)
- Previous Imaging (for comparison)
- Priority (Routine/Urgent/STAT)
- Special Instructions

5.6.2 PACS (Picture Archiving and Communication System)

Functionality:

- DICOM image storage
- Image viewer (web-based)
- Multi-planar reconstruction (MPR)
- 3D rendering
- Measurement tools
- Annotation tools
- Image comparison (side-by-side)
- CD burning (for patients)
- External sharing (secure link)

DICOM Features:

- DICOM Modality Worklist
- DICOM Storage
- DICOM Query/Retrieve
- DICOM Print
- HL7 integration

Image Viewer Tools:

- Zoom, Pan, Rotate
- Window/Level adjustment
- Magnifying glass
- Length/Angle measurement
- Region of Interest (ROI) measurement
- Cine loop (for series)
- Hanging protocols (custom layouts)

Storage:

- Short-term: SSD storage (6 months) for fast access
- Long-term: Nearline/Archive storage (7+ years)
- Backup: Offsite/cloud backup
- Compression: Lossless DICOM compression

5.6.3 Reporting & Structured Templates**Report Generation:**

- Template-based reporting
- Voice recognition (speech-to-text)
- Structured reporting (RADLEX)
- Free-text reporting
- Image annotation in report
- Comparison with prior studies
- Standardized terminology

Report Templates (Examples):

Chest X-Ray Template:

TECHNIQUE: PA and lateral chest radiographs

COMPARISON: [Prior study date or "None"]

FINDINGS:

- Lungs: [Normal/Abnormal]
- Heart: [Size normal/enlarged]
- Mediastinum: [Normal/Abnormal]
- Bones: [Normal/Abnormal]
- Soft tissues: [Normal/Abnormal]

IMPRESSION:

1. [Primary finding]
2. [Additional findings]

RECOMMENDATIONS:

- [Follow-up needed or not]

Reported by: [Radiologist Name]
Date & Time: [Auto-filled]

Features:

- Macro/shortcut phrases
- Auto-population from prior reports
- Critical findings flag
- Addendum functionality
- Peer review mechanism
- Turnaround time tracking
- Discrepancy reporting

Critical Results Notification:

- Pneumothorax
- Fracture with displacement
- Free intraperitoneal air
- Intracranial hemorrhage
- Pulmonary embolism
- Aortic dissection → Automatic SMS/call to ordering physician

User Stories:

US-RIS-001: As a radiologist, I want to view images from any location
US-RIS-002: As a radiologist, I want to compare current and prior images side-by-side
US-RIS-003: As a doctor, I want to access imaging reports within 30 minutes of scan
US-RIS-004: As a patient, I want to receive my imaging CD before discharge
US-RIS-005: As a technician, I want the worklist to update automatically

MODULE 7: Billing & Financial Management

5.7.1 Charge Capture

Functionality:

- Automatic charge capture from all modules
- Manual charge entry
- Service price list (tariff)
- Package/bundle pricing
- Discount management
- Charge reversal/adjustment

Billable Services:

Consultation Fees:

- General Practitioner: 5,000 XAF
- Specialist: 10,000 XAF
- Emergency: 15,000 XAF

Procedures:

- Minor surgery: 50,000 XAF
- Major surgery: 200,000-500,000 XAF
- Endoscopy: 75,000 XAF

Diagnostics:

- X-Ray: 8,000 XAF
- Ultrasound: 15,000 XAF
- CT Scan: 80,000 XAF
- MRI: 150,000 XAF
- CBC: 3,000 XAF
- Blood Chemistry: 5,000-15,000 XAF

Accommodation (per day):

- General Ward: 10,000 XAF
- Semi-private: 25,000 XAF
- Private Room: 50,000 XAF
- ICU: 100,000 XAF

Other:

- Medications: As per pharmacy prices
- Medical supplies: As per usage
- Ambulance: 20,000 XAF

Charge Posting:

- Real-time posting (as services delivered)
- Batch posting (end of day)
- Department-wise posting
- Package posting

5.7.2 Patient Billing**Bill Generation:****INVOICE**

Hospital Name: Regional Hospital of Malabo
Patient: Juan Nguema Mbomio
Patient ID: 2026-MLO-00542
Date: 22 May 2026

Services:

Description	Qty	Rate	Amount
-------------	-----	------	--------

Consultation (Dr. Ondo)	1	10,000	10,000 XAF
CBC Test	1	3,000	3,000 XAF
Chest X-Ray	1	8,000	8,000 XAF
Medications	-	-	5,500 XAF
Medical Supplies	-	-	2,000 XAF
SUBTOTAL			28,500 XAF
Discount (10%)			-2,850 XAF
TAX (if applicable)			0 XAF
TOTAL			25,650 XAF
Insurance Coverage (80%)			-20,520 XAF
PATIENT RESPONSIBILITY			5,130 XAF

Payment Method: Cash / Card / Insurance

Cashier: Maria Obiang

Receipt #: MLO-2026-0542

Features:

- Interim bills (for inpatients)
- Final bill at discharge
- Combined billing (multiple visits)
- Itemized vs summary view
- Multi-currency support
- Tax calculation
- Rounding rules

Payment Processing:

- Cash payment
- Card payment (POS integration)
- Mobile money (integration)
- Insurance claim submission
- Credit/Installment plans
- Advance/Deposit management
- Refund processing

Payment Plans:

- Down payment + installments
- Interest calculation
- Payment reminders
- Default tracking

5.7.3 Insurance Management

Insurance Provider Database:

- Insurance company details
- Coverage plans
- Co-pay/deductible amounts
- Pre-authorization requirements
- Claim submission process
- TAT (Turnaround Time) for reimbursement

Insurance Verification:

1. Patient provides insurance card
2. Staff enters policy number
3. System verifies eligibility (real-time API or manual)
4. Coverage details displayed:
 - Covered services
 - Co-pay amount
 - Deductible remaining
 - Maximum benefit
5. Pre-authorization check (if required)
6. Services rendered
7. Claim auto-generated
8. Claim submitted electronically

Claim Management:

- Claim generation
- Claim submission (electronic/paper)
- Claim tracking (submitted/pending/approved/rejected)
- Rejection management (resubmission)
- Payment posting
- Denial reasons tracking
- Appeal management

Government/Social Security:

- INSESO integration (for social security beneficiaries)
- Government employee coverage
- Exemption management (indigent care)
- Subsidy tracking

User Stories:

US-BL-001: As a cashier, I want to generate a bill in under 2 minutes

US-BL-002: As a patient, I want to know my out-of-pocket cost before treatment

US-BL-003: As a billing clerk, I want to verify insurance coverage in real-time

```
US-BL-004: As a finance manager, I want to see daily revenue by department
US-BL-005: As a patient, I want to pay via mobile money
```

5.7.4 Financial Reporting & Analytics

Standard Reports:

Daily Reports:

- Daily collection report (by cashier)
- Daily revenue by department
- Pending bills report
- Payment method breakdown

Monthly Reports:

- Monthly revenue report
- Outstanding receivables (aging)
- Insurance AR aging
- Bad debt report
- Revenue vs budget
- Department profitability

Annual Reports:

- Annual financial statements
- Audit reports
- Tax reports
- Government reporting

Analytics Dashboard:

- Revenue trends (daily/monthly/yearly)
- Top revenue-generating departments
- Payment method distribution
- Average bill value
- Collection efficiency
- Outstanding AR trends
- Payer mix (cash vs insurance vs government)

KPIs:

- Daily revenue
- Average revenue per patient (ARPOB)
- Collection rate
- AR days
- Claim denial rate
- Bad debt percentage

MODULE 8: Inpatient Management (IPD)

5.8.1 Admission Management

Admission Process:

1. Admission order from doctor
2. Bed availability check
3. Admission type selection:
 - Elective admission (planned)
 - Emergency admission
 - Maternity admission
 - Transfer from other facility
4. Room/bed assignment
5. Admission paperwork:
 - Consent forms
 - Rights and responsibilities
 - Advance directive
 - Insurance verification
6. Admission note by doctor
7. Nursing assessment
8. Care plan creation
9. Admission notification (bed management, pharmacy, dietary)

Admission Details:

- Admission Date & Time
- Admitting Doctor
- Admitting Diagnosis
- Department/Ward
- Room/Bed Number
- Admission Type (Medical/Surgical/Maternity/Pediatric)
- Admission Source (ER/OPD/Transfer)
- Expected Length of Stay
- Insurance/Payment Status

5.8.2 Bed Management

Bed Tracking:

Hospital Layout:

```
└─ Building
    └─ Floor
        └─ Ward/Department
            └─ Room
                └─ Bed
```

Bed Status:

- Available (clean and ready)
- Occupied

- Reserved
- Blocked (maintenance)
- Dirty (needs housekeeping)
- Under cleaning

Features:

- Real-time bed status dashboard
- Bed census (current occupancy)
- Bed turnaround time tracking
- Housekeeping integration
- Bed reservation system
- Transfer management (bed to bed)
- Bed capacity planning

Bed Census Report:

Regional Hospital of Malabo - Bed Census
Date: 22 May 2026, 08:00 AM

Ward	Total	Occupied	Available	Blocked
Medical	40	35	4	1
Surgical	50	42	7	1
ICU	10	8	2	0
Maternity	30	25	5	0
Pediatric	25	18	6	1
TOTAL	155	128	24	3

Occupancy Rate: 82.6%

Housekeeping Workflow:

1. Patient discharged
2. Bed status: "Dirty"
3. Housekeeping notified automatically
4. Housekeeper accepts task
5. Bed status: "Under cleaning"
6. Cleaning completed
7. Housekeeper marks complete
8. Bed status: "Available"
9. Average cleaning time: 30 minutes

5.8.3 Nursing Documentation

Nursing Assessments:

- Admission assessment
- Shift assessment (every 8/12 hours)
- Pain assessment
- Fall risk assessment
- Pressure ulcer risk assessment
- Nutrition screening

Vital Signs Charting:

- Temperature, Pulse, BP, RR, SpO2
- Fluid intake/output chart
- Glucose monitoring chart
- Pain score chart
- Graphical flowsheet (TPR chart)

Nursing Care Plan:

Problem: Risk for Falls

Goal: Patient will not experience falls during hospitalization

Interventions:

- Bed in low position
- Call bell within reach
- Hourly rounding
- Non-slip socks provided
- Bedside commode available

Evaluation: Patient remained fall-free

Medication Administration Record (MAR):

Patient: Juan Nguema

Date: 22 May 2026

Medication	Dose	Route	Time	Given By	Initials
Metformin	500mg	PO	08:00	M.Obiang	MO
			20:00	P.Ndong	PN
Lisinopril	10mg	PO	08:00	M.Obiang	MO

Administered: ☒ | Missed: ☐ | Refused: ☐

Features:

- Barcode medication administration (right patient, right drug, right dose)

- Missed dose documentation
- PRN (as needed) medication tracking
- Controlled substance tracking
- Adverse reaction reporting

Nursing Notes:

- SOAP format notes
- Critical event documentation
- Patient/family education
- Discharge planning notes

5.8.4 Discharge Management**Discharge Process:**

1. Doctor discharge order
2. Discharge summary creation
3. Discharge medications prescribed
4. Patient education (medications, follow-up, diet)
5. Follow-up appointment scheduled
6. Final billing
7. Discharge clearance
8. Discharge instructions printed
9. Patient signature (acknowledgment)
10. Bed status updated
11. Discharge notification (pharmacy, dietary, etc.)

Discharge Summary:**DISCHARGE SUMMARY**

Patient: Juan Nguema Mbomio
ID: 2026-MLO-00542
Admission Date: 15 May 2026
Discharge Date: 22 May 2026
Length of Stay: 7 days

ADMITTING DIAGNOSIS:

Acute Appendicitis

HOSPITAL COURSE:

Patient admitted with acute appendicitis. Underwent laparoscopic appendectomy on 15 May 2026 without complications. Post-operative recovery uneventful.

PROCEDURES PERFORMED:

- Laparoscopic Appendectomy (15 May 2026)

DISCHARGE DIAGNOSIS:

1. Acute Appendicitis (post-operative)
2. Type 2 Diabetes Mellitus

DISCHARGE MEDICATIONS:

1. Metformin 500mg PO twice daily
2. Paracetamol 500mg PO as needed for pain
3. Amoxicillin 500mg PO three times daily x 5 days

DISCHARGE CONDITION: Stable

FOLLOW-UP:

- Wound check in 7 days (29 May 2026)
- Dr. Ondo, Surgical OPD

PATIENT EDUCATION:

- Wound care instructions provided
- Activity restrictions explained
- Warning signs discussed

Prepared by: Dr. Milagrosa Ondo

Date: 22 May 2026

Discharge Instructions:

- Medication list with instructions
- Activity restrictions
- Diet recommendations
- Wound care (if applicable)
- Warning signs (when to seek help)
- Follow-up appointments
- Medical certificate (if needed)

MODULE 9: Emergency Department (ER/A&E)**5.9.1 Triage System****Triage Levels (ESI - Emergency Severity Index):**

Level 1 - RESUSCITATION (Red)

Life-threatening, immediate

Examples: Cardiac arrest, severe trauma, unconscious

Response time: Immediate

Level 2 - EMERGENT (Orange)

High risk, very urgent

Examples: Chest pain, severe bleeding, fractures

Response time: <10 minutes

Level 3 - URGENT (Yellow)
Moderate risk, urgent
Examples: Moderate pain, minor fractures, vomiting
Response time: <30 minutes

Level 4 - LESS URGENT (Green)
Low risk, semi-urgent
Examples: Minor injuries, mild pain
Response time: <60 minutes

Level 5 - NON-URGENT (Blue)
Non-urgent
Examples: Prescription refills, chronic complaints
Response time: <120 minutes

Triage Assessment:

- Chief Complaint
- Vital Signs
- Pain Score (0-10)
- Mechanism of Injury (if trauma)
- Allergies
- Current Medications
- Medical History (brief)
- Last Tetanus Shot (if trauma)
- Pregnancy Status (females)
- Triage Level Assignment
- Triage Nurse Signature
- Time Stamped

Features:

- Color-coded triage tags
- Digital triage board (TV display)
- Auto-escalation (if waiting too long)
- Re-triage mechanism
- Mass casualty incident mode

5.9.2 Emergency Room Management

ER Dashboard:

Emergency Department - Real-time Dashboard

Total Patients: 15

|— Waiting: 8

|— In Treatment: 5

|— Awaiting Admission: 2

By Triage Level:

Red:  2

Orange:  4

Yellow:  6

Green:  3

Average Wait Time: 25 minutes

Longest Wait: 45 minutes (Patient #2345)

Bed Status:

Trauma Bays: 2/3 occupied

Resuscitation: 1/2 occupied

Fast Track: 4/6 occupied

Features:

- Patient tracking board
- Room assignment
- Doctor assignment
- Wait time monitoring
- Alerts for prolonged wait
- LWBS tracking (Left Without Being Seen)
- Ambulance arrival notifications

5.9.3 Emergency Documentation

ER Note Template:

EMERGENCY DEPARTMENT NOTE

Time In: 14:35

Time Seen: 14:42

Triage Level: 2 (Orange)

CHIEF COMPLAINT:

Chest pain

HPI (History of Present Illness):

45 y/o male with sudden onset chest pain...

VITAL SIGNS:

BP: 145/92, HR: 98, RR: 18, Temp: 37.2°C, SpO2: 97%

PHYSICAL EXAMINATION:

...

TESTS/IMAGING ORDERED:

- ECG (completed 14:50) - Normal
- Cardiac enzymes (pending)

- Chest X-ray (completed 15:00) - Clear

DIAGNOSIS:

Atypical chest pain, rule out cardiac

TREATMENT PROVIDED:

- Aspirin 300mg PO given
- IV access established
- Cardiac monitoring

DISPOSITION:

Admit to Cardiology for observation

Time Out: 15:30

Total ED Time: 55 minutes

Attending: Dr. Essono

Fast Track (Minor Injuries):

- Quick registration
- Simplified documentation
- Rapid discharge
- Common conditions: lacerations, sprains, minor burns

5.9.4 Ambulance Integration

Ambulance Dispatch:

- Call received (time stamped)
- Ambulance assigned
- GPS tracking
- ETA to scene
- ETA to hospital
- Pre-arrival notification to ER

Pre-Hospital Data:

- Patient demographics (if known)
- Vital signs
- Mechanism of injury
- Treatment provided in ambulance
- Transmitted to hospital en route

MODULE 10: Operation Theatre (OT) Management

5.10.1 Surgery Scheduling

OT Booking:

Surgery Schedule - 23 May 2026

OT 1 (General Surgery):

08:00-10:00 | Appendectomy | Dr. Ondo | Patient: Maria A.

10:30-12:00 | Hernia Repair | Dr. Ondo | Patient: Jose M.

14:00-16:00 | Cholecystectomy | Dr. Bindang | Patient: Ana N.

OT 2 (Orthopedics):

08:00-11:00 | Hip Replacement | Dr. Nsue | Patient: Pedro O.

12:00-14:00 | Fracture Fixation | Dr. Nsue | Patient: Carmen E.

OT 3 (Emergency):

AVAILABLE (reserved for emergencies)

Booking Details:

- Patient Information
- Surgeon & Anesthesiologist
- Procedure Name & Code
- Expected Duration
- Anesthesia Type (General/Spinal/Local)
- OT Room Assignment
- Special Equipment Needed
- Blood Arranged? (Y/N)
- Pre-op Checklist Complete
- Consent Signed

Features:

- Drag-and-drop scheduling
- OT utilization tracking
- Conflict detection
- Equipment availability check
- Team availability check
- Emergency slot management
- OT turnover time tracking

5.10.2 Pre-Operative Assessment

Pre-Anesthesia Checklist:

- ☐ Medical history reviewed
- ☐ Current medications documented
- ☐ Allergies confirmed
- ☐ NPO status confirmed (nothing by mouth)
- ☐ Pre-op labs reviewed (CBC, PT/PTT, etc.)
- ☐ ECG reviewed (if >40 years)

- ☐ Chest X-ray reviewed (if indicated)
- ☐ ASA classification assigned
- ☐ Anesthesia plan documented
- ☐ Informed consent signed
- ☐ IV access established
- ☐ Pre-medication given
- ☐ Site marking confirmed

ASA Classification:

- ASA I: Healthy patient
- ASA II: Mild systemic disease
- ASA III: Severe systemic disease
- ASA IV: Life-threatening disease
- ASA V: Moribund patient
- ASA VI: Brain-dead organ donor

5.10.3 Intra-Operative Documentation

OT Register/Log:

OPERATION THEATRE RECORD

Patient: Juan Nguema Mbomio
Procedure: Laparoscopic Appendectomy
Surgeon: Dr. Milagrosa Ondo
Assistant: Dr. Carlos Bindang
Anesthesiologist: Dr. Ana Nsue
Scrub Nurse: Nurse Maria Obiang
Circulating Nurse: Nurse Pedro Ela

Time In OT: 08:05
Anesthesia Start: 08:15
Surgery Start: 08:30
Surgery End: 09:45
Anesthesia End: 09:55
Time Out OT: 10:00
Total Time: 1 hour 55 minutes

Anesthesia Type: General Endotracheal
Anesthesia Drugs: Propofol, Fentanyl, Rocuronium

Procedure Performed:
Laparoscopic appendectomy performed. Inflamed appendix identified and removed. No complications.

Specimens Sent:
Appendix → Histopathology

Estimated Blood Loss: 50 ml
Fluids Given: Ringers Lactate 1000ml

Blood Transfused: None

Implants Used: None

Complications: None

Condition at End: Stable

Destination: Recovery Room

WHO Surgical Safety Checklist:

SIGN IN (Before Anesthesia):

- ☒ Patient identity confirmed
- ☒ Site marked
- ☒ Anesthesia safety check complete
- ☒ Pulse oximeter functioning

TIME OUT (Before Skin Incision):

- ☒ Team introduction
- ☒ Confirm patient, site, procedure
- ☒ Anticipated critical events discussed
- ☒ Antibiotic prophylaxis given
- ☒ Essential imaging displayed

SIGN OUT (Before Patient Leaves OT):

- ☒ Procedure recorded correctly
- ☒ Instrument/sponge count correct
- ☒ Specimens labeled
- ☒ Equipment problems addressed
- ☒ Key concerns for recovery/ward

Features:

- Real-time OT status board
- Anesthesia charting
- Fluid balance tracking
- Medication administration
- Critical events logging
- Image capture (if available)

5.10.4 Post-Operative Care

Recovery Room (PACU):

Post-Anesthesia Recovery Score (Aldrete Score):

- Activity (movement): 0-2
- Respiration: 0-2

- Circulation: 0-2
- Consciousness: 0-2
- SpO2: 0-2
Total: /10 (≥9 for discharge)

Monitoring Frequency: Every 15 minutes

Discharge Criteria Met:

- ☒ Aldrete score ≥9
- ☒ Stable vital signs
- ☒ No active bleeding
- ☒ Pain controlled
- ☒ No nausea/vomiting
- ☒ Able to ambulate (if appropriate)

Discharged to: General Ward / ICU / Home

Post-Op Orders:

- Transfer to: Medical Ward, Room 304, Bed 2
- Vital signs: Every 4 hours
- Activity: Bed rest for 6 hours, then ambulate
- Diet: Clear liquids, advance as tolerated
- IV Fluids: NS 100ml/hr x 24 hours
- Medications:
 - Paracetamol 1g IV every 8 hours PRN pain
 - Ondansetron 4mg IV every 8 hours PRN nausea
- Drain care: Monitor output every shift
- Follow-up: Surgical OPD in 7 days

MODULE 11: Maternity & Obstetrics

5.11.1 Antenatal Care (ANC)

Antenatal Registration:

- LMP (Last Menstrual Period)
- EDD (Expected Date of Delivery) - auto-calculated
- Gravida/Para status (G2P1: 2nd pregnancy, 1 birth)
- Obstetric history (previous pregnancies)
- Medical history
- Surgical history
- Family history
- Allergies
- Blood group & Rh factor
- Booking visit (1st ANC visit)

ANC Schedule (WHO Recommendations):

Visit 1: 8-12 weeks (Booking visit)
Visit 2: 20 weeks
Visit 3: 26 weeks
Visit 4: 30 weeks
Visit 5: 34 weeks
Visit 6: 36 weeks
Visit 7: 38 weeks
Visit 8: 40 weeks
(Weekly thereafter if not delivered)

ANC Visit Documentation:

Gestational Age: 28 weeks
Weight: 68 kg (+2 kg since last visit)
Blood Pressure: 120/75
Fundal Height: 27 cm
Fetal Heart Rate: 145 bpm
Fetal Movement: Adequate
Presentation: Cephalic
Edema: None
Proteinuria: Negative

Investigations:

- Hemoglobin: 11.2 g/dL
- Blood Glucose: 92 mg/dL

Supplements:

- Iron + Folic Acid
- Calcium

Vaccinations:

- TT1 given (15 weeks)
- TT2 given (20 weeks)

Next Visit: 30 weeks
High-risk factors: None

High-Risk Pregnancy Flags:

- Age <18 or >35 years
- Previous C-section
- Multiple pregnancy (twins/triplets)
- Hypertension/Pre-eclampsia
- Diabetes
- Previous preterm delivery
- Previous stillbirth

- Rh-negative mother

Features:

- ANC appointment reminders
- Growth chart (fundal height)
- Vaccination schedule
- Educational content (prenatal nutrition, danger signs)
- Birth plan documentation

5.11.2 Labor & Delivery

Admission to Labor Ward:

Reason for Admission:

- ☒ In labor (contractions)
- ☐ Induction of labor
- ☐ Scheduled C-section
- ☐ Complications

Admission Assessment:

- Gestational Age: 39 weeks 3 days
- Contractions: Every 5 minutes, lasting 45 seconds
- Membrane Status: Intact / Ruptured (time: ____)
- Cervical Dilation: 4 cm
- Cervical Effacement: 60%
- Station: -1
- Fetal Heart Rate: 140 bpm
- Color of Liquor: Clear / Meconium stained

Partograph (Labor Monitoring Chart):

Progress of Labor:

Time: 14:00 15:00 16:00 17:00 18:00

Dilation: 4 5 6 8 10 cm

Station: -1 0 +1 +2 +3

FHR: 140 145 138 142 150 bpm

Contractions: // /// //// //// /////

Alert Line: [Graphical line]

Action Line: [Graphical line]

If cervical dilation crosses action line → Consider intervention

Delivery Documentation:

DELIVERY NOTE

Date: 22 May 2026

Time of Delivery: 18:35

Mode of Delivery:

- ☒ Spontaneous Vaginal Delivery
☐ Assisted Delivery (Vacuum/Forceps)
☐ Cesarean Section

Indication (if operative): N/A

Baby Details:

Sex: Male

Weight: 3.2 kg

Length: 50 cm

Head Circumference: 35 cm

APGAR Score: 1 min = 9, 5 min = 10

Cried immediately: Yes

Birth defects: None

Placenta:

Delivered: Complete

Weight: 550 g

Membrane: Complete

Cord vessels: 3 (normal)

Blood Loss: 350 ml (normal)

Perineal Tear: 1st degree (repaired)

Mother's Condition: Stable

Baby's Condition: Good

Delivered by: Dr. Ana Nsue, Obstetrician

APGAR Score:

Parameter	0 points	1 point	2 points
Appearance (color)	Blue	Extremities blue	Pink
Pulse	Absent	<100 bpm	>100 bpm
Grimace (reflexes)	None	Weak cry	Strong cry
Activity (tone)	Limp	Some flexion	Active
Respiration	Absent	Weak	Strong

Score: 7-10 = Normal, 4-6 = Moderate distress, 0-3 = Severe distress

5.11.3 Postnatal Care

Mother's Postpartum Care:

Vital signs monitoring:

- Every 15 min x 1 hour
- Every 30 min x 2 hours
- Every 4 hours x 24 hours
- Then every shift

Assessment:

- Uterine involution (fundus check)
- Lochia (vaginal discharge) - color, amount
- Perineal wound check
- Breast examination
- Breastfeeding assessment

Medications:

- Pain relief (Paracetamol)
- Iron + Folic acid supplementation
- Vitamin A (single dose within 24 hours)

Discharge Planning:

- Family planning counseling
- Breastfeeding education
- Postnatal follow-up (6 weeks)
- Danger signs education

Newborn Care:

Immediate Care:

- Vitamin K injection (within 1 hour)
- Eye prophylaxis (Tetracycline eye ointment)
- Hepatitis B vaccine (within 24 hours)
- BCG vaccine
- Birth weight recorded

Newborn Examination:

- General appearance
- Skin color (jaundice check)
- Head (fontanelles, molding)
- Eyes (red reflex)
- Heart & lungs
- Abdomen (cord stump)
- Genitalia
- Hips (developmental dysplasia screening)
- Spine
- Reflexes (Moro, sucking, grasp)

Feeding:

- Breastfeeding initiated within 1 hour
- Exclusive breastfeeding encouraged

Discharge Criteria (Mother & Baby):

- Vital signs stable x 24 hours
- No excessive bleeding
- Baby feeding well
- Passed urine & meconium
- Cord stump dry
- Jaundice assessment (if present)
- Immunization card provided
- Follow-up appointment scheduled

MODULE 12: Pediatrics

5.12.1 Growth Monitoring

Growth Charts (WHO Standards):

- Weight-for-age (birth to 5 years)
- Length/Height-for-age
- Weight-for-length/height
- BMI-for-age (2-19 years)
- Head circumference-for-age (birth to 5 years)

Features:

- Automatic plotting on growth charts
- Z-score calculation
- Growth velocity tracking
- Malnutrition screening (underweight, wasting, stunting)
- Overweight/obesity detection
- Growth trend alerts

Nutritional Status Classification:

Severe Acute Malnutrition (SAM): Weight-for-height < -3 SD
Moderate Acute Malnutrition (MAM): Weight-for-height -3 to -2 SD
Normal: Weight-for-height -2 to $+2$ SD
Overweight: Weight-for-height $> +2$ SD
Obese: Weight-for-height $> +3$ SD

5.12.2 Immunization Tracking

National Immunization Schedule:

At Birth:

- BCG
- OPV-0 (Oral Polio Vaccine)

- Hepatitis B-1

6 Weeks:

- OPV-1
- Pentavalent-1 (DPT + HepB + Hib)
- Pneumococcal-1
- Rotavirus-1

10 Weeks:

- OPV-2
- Pentavalent-2
- Pneumococcal-2
- Rotavirus-2

14 Weeks:

- OPV-3
- Pentavalent-3
- Pneumococcal-3

9 Months:

- Measles-Rubella-1
- Yellow Fever
- Vitamin A

18 Months:

- Measles-Rubella-2
- OPV Booster

5 Years:

- DPT Booster

Features:

- Automatic schedule generation based on birth date
- Missed vaccination alerts
- Batch tracking (vaccine lot number, expiry date)
- Cold chain temperature monitoring
- Adverse event reporting (AEFI)
- Immunization card printing
- Coverage reporting

Vaccination Record:

Patient: Baby Maria Obiang
Date of Birth: 15 March 2026

Vaccine	Date Given	Batch No.	Given By
BCG	15/03/2026	BCG-2026-05	M.Ela
OPV-0	15/03/2026	OPV-2026-12	M.Ela

HepB-1	15/03/2026	HEP-2026-08	M.Ela
OPV-1	26/04/2026	OPV-2026-12	P.Nsue
Pentavalent-1	26/04/2026	PNT-2026-03	P.Nsue

Next Due: 24 May 2026 (OPV-2, Pentavalent-2)

5.12.3 Pediatric Dosing Calculator

Features:

- Weight-based dosing
- Age-based dosing
- BSA-based dosing (body surface area)
- Maximum dose warnings
- Dosing schedules

Example:

Drug: Paracetamol
Patient Weight: 15 kg
Dose: 15 mg/kg/dose
Calculated: 225 mg per dose
Available: 120mg/5ml syrup
Volume: 9.4 ml per dose
Frequency: Every 6 hours
Maximum daily dose: 900 mg (not exceeded ✓)

MODULE 13: Pharmacy - Advanced Features

5.13.1 Drug Information Database

Comprehensive Drug Information:

Drug Name: Amoxicillin
Generic Name: Amoxicillin Trihydrate
Brand Names: Amoxil, Trimox

Therapeutic Class: Antibiotic (Penicillin)
Pharmacologic Class: Beta-lactam antibiotic

Indications:
- Respiratory tract infections
- Urinary tract infections
- Skin infections
- Otitis media

Dosage:

- Adults: 250-500 mg every 8 hours
- Children: 20-40 mg/kg/day divided every 8 hours

Contraindications:

- Penicillin allergy
- History of severe hypersensitivity

Side Effects:

- Common: Nausea, diarrhea, rash
- Serious: Anaphylaxis, C. difficile colitis

Drug Interactions:

- Warfarin (increased INR)
- Oral contraceptives (decreased effectiveness)

Pregnancy Category: B (safe)

Lactation: Compatible with breastfeeding

Storage: Room temperature, protect from moisture

Clinical Decision Support:

- Allergy checking
- Drug-drug interactions
- Drug-disease interactions
- Renal dose adjustment recommendations
- Pregnancy/lactation safety
- Geriatric dosing considerations

5.13.2 Controlled Substances Management**Schedule/Classification:**

- Schedule I: High abuse potential, no medical use (e.g., Heroin)
- Schedule II: High abuse potential (e.g., Morphine, Fentanyl)
- Schedule III: Moderate abuse potential (e.g., Codeine combinations)
- Schedule IV: Low abuse potential (e.g., Benzodiazepines)
- Schedule V: Lowest abuse potential (e.g., Cough syrups with codeine)

Features:

- Separate storage tracking
- Dual signature requirement (nurse + witness)
- Wastage documentation
- Audit trail
- Regulatory reporting
- Prescription limits (refill restrictions)

Controlled Substance Log:

Drug: Morphine 10mg Injection
 Opening Balance (1 May): 50 ampoules

Date	Qty	Patient ID	Doctor	Nurse	Witness	Balance
02/05	-1	2026-542	Ondo	Obiang	Nsue	49
03/05	-2	2026-589	Bindang	Ela	Mbomio	47
05/05	+20	[Received]	-	Obiang	Manager	67
...						

Current Balance: 43 ampoules
 Physical Count: 43 ampoules ✓ (Verified 20/05/2026)

5.13.3 Adverse Drug Reaction (ADR) Reporting

ADR Form:

Patient: [Name, ID, Age, Gender]
 Date of Reaction: [Date]
 Suspected Drug(s): [Drug name, dose, route, dates]
 Reaction Description: [Free text]
 Severity:
☐ Mild (no intervention needed)
☐ Moderate (intervention required)
☐ Severe (life-threatening)
 Seriousness:
☐ Death
☐ Life-threatening
☐ Hospitalization (initial or prolonged)
☐ Disability
☐ Congenital anomaly
 Outcome:
☐ Recovered
☐ Recovering
☐ Not recovered
☐ Fatal
☐ Unknown
 Reporter: [Name, Designation]

Features:

- Online ADR reporting
- Pharmacovigilance database
- Signal detection (pattern recognition)
- Regulatory body notification

- Patient notification
 - Formulary review based on ADRs
-

MODULE 14: Telemedicine (Future Module)

5.14.1 Virtual Consultations

Features:

- Video consultation (doctor-patient)
- Audio consultation
- Chat consultation
- Screen sharing
- Digital whiteboard
- E-prescription at end of consultation
- Payment integration

Use Cases:

- Follow-up consultations
- Chronic disease management
- Mental health counseling
- Specialist consultations (remote areas)
- Second opinions

5.14.2 Remote Monitoring

Features:

- Integration with wearable devices
 - Blood pressure monitoring
 - Blood glucose monitoring
 - Weight tracking
 - Heart rate monitoring
 - Automatic alerts for abnormal values
-

MODULE 15: Reporting & Analytics

5.15.1 Operational Reports

Daily Reports:

- Daily Patient Census
- Daily Revenue Report
- Daily Admission/Discharge Report
- Daily Bed Occupancy
- Daily Laboratory Tests

- Daily Pharmacy Dispensing
- Daily Emergency Department Activity

Weekly Reports:

- Weekly KPI Dashboard
- Weekly Appointment Statistics
- Weekly Infection Control Report
- Weekly Medication Errors

Monthly Reports:

- Monthly Patient Statistics
- Monthly Revenue Analysis
- Monthly Department Performance
- Monthly HR Report (attendance, leave)
- Monthly Inventory Report
- Monthly Quality Indicators

5.15.2 Ministry Dashboard

Strategic KPIs for Ministry:

National Health Dashboard

Total Patients Treated (This Month): 45,234

└ Malabo Regional: 15,890

└ Bata Regional: 12,456

└ Sampaka: 8,234

└ INSESO Bata: 5,678

└ Dr. Ioeri Polyclinic: 2,976

Total Revenue: 1,234,567,890 XAF

Top 10 Diagnoses:

1. Malaria: 23% (10,404 cases)

2. Hypertension: 15% (6,785 cases)

3. Diabetes: 8% (3,619 cases)

...

Bed Occupancy Rate: 78.4% (national average)

Average Length of Stay: 4.2 days

Patient Satisfaction: 87.3%

Staff Satisfaction: 82.1%

Emergency Response Time: 12.3 minutes (average)

Medication Stock-Outs: 3 items (national)

- Amoxicillin 500mg (Sampaka)
- Insulin (Bata)
- Paracetamol IV (INSESO)

Geographic Analysis:

- Heat map of disease distribution
- Facility utilization by region
- Resource allocation recommendations

Predictive Analytics:

- Disease outbreak prediction
- Bed demand forecasting
- Medication demand forecasting
- Budget forecasting

5.15.3 Public Health Surveillance

Notifiable Diseases Tracking:

Immediate Notification (within 24 hours):

- Cholera
- Ebola
- Yellow Fever
- Meningitis
- Measles
- COVID-19 / Pandemic diseases

Weekly Notification:

- Malaria
- Tuberculosis
- HIV/AIDS
- Sexually Transmitted Infections

Features:

- Automatic case detection
- Line listing generation
- Epidemic curve visualization
- Contact tracing module
- GIS mapping
- Alert notifications to Ministry

Example Alert:

 OUTBREAK ALERT

Disease: Measles
Location: Bata Region
Cases in last 7 days: 15
(Threshold: 5 cases)

- Action Required:
- Enhanced surveillance
 - Vaccination campaign
 - Public health advisory

Generated: 22 May 2026, 09:00

6. User Roles & Permissions

6.1 Role-Based Access Control (RBAC)

Role Hierarchy:

```
System Administrator
├── Ministry Officials
├── Hospital Director
│   ├── Medical Director
│   ├── Nursing Director
│   ├── Finance Manager
│   └── IT Manager
├── Department Heads
│   ├── Emergency Department Head
│   ├── Laboratory Manager
│   ├── Radiology Manager
│   └── Pharmacy Manager
├── Clinical Staff
│   ├── Doctors (Specialists, GPs)
│   ├── Nurses (Senior, Junior)
│   ├── Midwives
│   └── Allied Health (Physiotherapist, etc.)
├── Support Staff
│   ├── Receptionists
│   ├── Billing Clerks
│   ├── Pharmacists
│   ├── Lab Technicians
│   ├── Radiology Technicians
│   └── Medical Records Staff
└── External Users
    ├── Patients (Portal)
    └── Insurance Companies
```

6.2 Permission Matrix

Example: Doctor Permissions

Module	Create	Read	Update	Delete	Print
Patient Registration	✓	✓	✓	✗	✓
Medical Records	✓	✓	✓	✗	✓
Appointments	✓	✓	✓	✓	✓
Prescriptions	✓	✓	✓	✓	✓
Lab Orders	✓	✓	✓	✓	✓
Imaging Orders	✓	✓	✓	✓	✓
Admission/Discharge	✓	✓	✓	✗	✓
Billing	✗	✓	✗	✗	✓
Financial Reports	✗	✗	✗	✗	✗
Staff Management	✗	✓	✗	✗	✗
System Configuration	✗	✗	✗	✗	✗

Example: Receptionist Permissions

Module	Create	Read	Update	Delete	Print
Patient Registration	✓	✓	✓	✗	✓
Medical Records	✗	✓	✗	✗	✗
Appointments	✓	✓	✓	✓	✓
Prescriptions	✗	✗	✗	✗	✓
Lab Orders	✗	✓	✗	✗	✓
Imaging Orders	✗	✓	✗	✗	✓
Admission/Discharge	✓	✓	✓	✗	✓
Billing	✓	✓	✓	✗	✓
Financial Reports	✗	✗	✗	✗	✗
Staff Management	✗	✗	✗	✗	✗
System Configuration	✗	✗	✗	✗	✗

6.3 Data Access Restrictions

Patient Data Privacy:

- Staff can only access patients under their care
- Break-the-glass mechanism (emergency access with audit)
- VIP patient protection (additional authorization required)
- Deceased patient records (restricted access)

Department-Level Restrictions:

- Lab staff: access only to lab module
- Pharmacy staff: access only to pharmacy module
- Cross-department access requires explicit permission

Facility-Level Restrictions:

- Staff normally access only their facility's data
 - Ministry officials: access to all facilities
 - Specialists: access to multiple facilities (if they work at multiple locations)
-

7. Technical Requirements

7.1 Hardware Requirements

Server Infrastructure (Cloud or On-Premise):**Application Servers:**

- CPU: 16 cores minimum
- RAM: 64 GB minimum
- Storage: 500 GB SSD (OS & applications)
- Network: 10 Gbps NIC

Database Servers:

- CPU: 32 cores minimum
- RAM: 128 GB minimum (database workloads are memory-intensive)
- Storage: 2 TB NVMe SSD (hot data), 10 TB HDD (warm data)
- RAID configuration for redundancy

Storage/Backup:

- Primary Storage: 50 TB (for 5 years of images, documents)
- Backup Storage: 50 TB (incremental backups)
- Archive Storage: 100 TB (cold storage for old records)

Local Servers (Per Hospital):

- CPU: 8 cores
- RAM: 32 GB
- Storage: 2 TB SSD
- Purpose: Offline mode, local caching

Client Devices:**Desktop Workstations:**

- CPU: Intel i5 or equivalent
- RAM: 8 GB minimum
- Storage: 256 GB SSD
- Display: 21-24 inch monitors (dual monitor setup for doctors/radiologists)
- Peripherals: Keyboard, mouse, barcode scanner

Tablets (for ward rounds):

- 10-12 inch tablets
- Android or iOS
- 4 GB RAM minimum
- LTE connectivity

Barcode Scanners:

- Handheld USB/Wireless scanners
- 2D barcode support (QR codes)

Label Printers:

- Thermal label printers (for patient wristbands, specimen labels)
- 300 DPI minimum

Receipt Printers:

- Thermal receipt printers (for billing)
- 80mm width

Medical Device Integration:

- Vital signs monitors (BP, SpO2, Thermometer) with HL7/USB export
- Laboratory analyzers (Hematology, Chemistry) with LIS interface
- ECG machines with digital export
- Ultrasound/X-Ray/CT/MRI machines with DICOM export

7.2 Software Requirements

Operating Systems:

- Servers: Linux (Ubuntu Server 22.04 LTS or CentOS 8)
- Desktops: Windows 10/11 or Linux
- Mobile: Android 10+, iOS 14+

Web Browsers:

- Chrome 100+
- Firefox 100+
- Safari 15+ (for Mac users)
- Edge 100+

Database:

- PostgreSQL 14+ (primary relational database)
- MongoDB 5+ (document storage)
- Redis 6+ (caching)

Development Frameworks:

- Backend: Node.js 18 LTS / Python 3.10+
- Frontend: React 18+ / Vue 3+

- Mobile: React Native 0.70+

APIs & Standards:

- REST API (primary)
- GraphQL (for complex queries)
- HL7 v2.5 (lab/radiology interfaces)
- FHIR (Fast Healthcare Interoperability Resources) - future
- DICOM (medical imaging)
- ICD-10 (diagnosis codes)
- LOINC (lab test codes)

Security:

- SSL/TLS 1.3 (all communications)
- AES-256 encryption (data at rest)
- JWT tokens (authentication)
- OAuth 2.0 (third-party integrations)
- Two-factor authentication (2FA) for privileged accounts

7.3 Network Requirements

Internet Connectivity:

- Malabo Regional Hospital: 100 Mbps dedicated fiber
- Bata Regional Hospital: 100 Mbps dedicated fiber
- Other facilities: 50 Mbps minimum
- Backup: 4G/5G cellular with failover

Internal Network:

- Gigabit Ethernet (1 Gbps) within each hospital
- WiFi 6 (802.11ax) for mobile devices
- Separate VLANs: Clinical, Administrative, Guest

VPN:

- Site-to-site VPN between hospitals and data center
- IPSec or OpenVPN
- Encryption: AES-256

Firewall:

- Application-level firewall
- Intrusion Detection/Prevention System (IDS/IPS)
- DDoS protection

Uptime Requirements:

- System Availability: 99.9% (less than 9 hours downtime per year)
- Planned maintenance windows: Monthly, off-peak hours

8. Integration Requirements

8.1 Internal Integrations

Module Interconnections:

```
Patient Registration
↓
├→ Appointment Scheduling
├→ Billing (demographics)
└→ EMR (medical records)

Doctor Orders
↓
├→ Laboratory (lab orders)
├→ Radiology (imaging orders)
├→ Pharmacy (prescriptions)
└→ Billing (charge capture)

Laboratory Results
↓
├→ EMR (results displayed)
└→ Doctor Notification (critical results)

Admission
↓
├→ Bed Management
├→ Pharmacy (inpatient medications)
├→ Dietary (meal orders)
├→ Billing (room charges)
└→ Nursing (care plans)

Discharge
↓
├→ Billing (final bill)
├→ Pharmacy (discharge medications)
├→ Bed Management (bed freed)
└→ Appointment Scheduling (follow-up)
```

Event-Driven Architecture:

- When lab result is critical → Trigger SMS to doctor
- When bed is freed → Notify housekeeping
- When prescription is written → Notify pharmacy
- When patient is admitted → Update bed census

8.2 External Integrations

Government Systems:

- Ministry of Health reporting portal

- National Health Insurance (if applicable)
- INSESO (Social Security) - eligibility verification
- Civil Registry (patient verification)
- National Drug Regulatory Authority (medication tracking)

Payment Gateways:

- Mobile Money providers (MTN Money, Orange Money, Airtel Money)
- Credit card processors
- Bank transfers

Laboratory Equipment:

- Hematology analyzers (Sysmex, Beckman Coulter)
- Chemistry analyzers (Roche, Abbott)
- Microbiology systems
- Blood gas analyzers

Radiology Equipment:

- DICOM interface with X-Ray, CT, MRI, Ultrasound machines
- PACS integration

Third-Party Services:

- SMS gateway (Twilio, Africa's Talking)
- Email service (SendGrid, AWS SES)
- Cloud storage (AWS S3, Azure Blob)
- Backup services

APIs to Expose:

- Patient Portal API (for mobile app)
- Insurance Verification API
- Appointment Booking API (for website)
- Telemedicine API

9. Security & Compliance

9.1 Data Security

Access Control:

- Role-based access control (RBAC)
- Principle of least privilege
- Multi-factor authentication (2FA) for admins
- Session timeout (15 minutes of inactivity)
- Password policy:
 - Minimum 10 characters
 - Mix of uppercase, lowercase, numbers, symbols

- Password expiry: 90 days
- No password reuse (last 5 passwords)

Data Encryption:

- Data at rest: AES-256 encryption (database, file storage)
- Data in transit: TLS 1.3 (all API calls, web traffic)
- Encrypted backups
- Secure key management (HSM or KMS)

Audit Logging:

- All user actions logged (who, what, when, where, why)
- Immutable audit logs
- Patient record access logging (who viewed which patient)
- Failed login attempt tracking (brute force detection)
- Log retention: 7 years

Network Security:

- Firewall (allow only necessary ports)
- Intrusion Detection/Prevention
- VPN for remote access
- Network segmentation (VLANs)
- Regular security scanning (vulnerability assessments)

Physical Security:

- Server room access control (biometric/card)
- CCTV surveillance
- Backup power (UPS, generators)
- Environmental monitoring (temperature, humidity)

9.2 Compliance & Standards

Healthcare Standards:

- HL7 v2.5 / FHIR (interoperability)
- DICOM (medical imaging)
- ICD-10 (diagnosis coding)
- LOINC (lab test coding)
- SNOMED CT (clinical terminology) - optional

Data Protection:

- Patient consent management
- Right to access (patients can request their records)
- Right to correction (patients can request corrections)
- Data retention policy (medical records: 20+ years)
- Data disposal policy (secure deletion after retention period)

Regulatory Compliance:

- Equatorial Guinea healthcare regulations
- WHO guidelines for digital health
- GDPR principles (even if not in EU, good practice)

Quality Certifications (Target):

- ISO 27001 (Information Security Management)
- ISO 13485 (Medical Device Software)
- HIPAA-like compliance (patient privacy)

9.3 Disaster Recovery & Business Continuity

Backup Strategy:

```
Level 1: Continuous
- Database transaction logs: Real-time replication

Level 2: Frequent
- Full database backup: Every 6 hours
- Incremental backup: Every 1 hour

Level 3: Daily
- Full system backup: Daily at 2 AM
- Application files, configurations

Level 4: Weekly
- Full backup to tape/offline storage: Every Sunday

Level 5: Monthly
- Archive backup (off-site): Every 1st of month
```

Backup Locations:

- Primary: Same data center (different storage)
- Secondary: Different data center in same country
- Tertiary: Cloud backup (AWS/Azure)
- Quaternary: Tape backup (off-site secure facility)

Recovery Time Objective (RTO):

- Critical systems (EMR, ER, OT): < 1 hour
- Important systems (Billing, Lab, Radiology): < 4 hours
- Non-critical systems (Reporting): < 24 hours

Recovery Point Objective (RPO):

- Critical data (patient records): < 5 minutes of data loss acceptable
- Important data (billing, lab results): < 1 hour

- Non-critical data: < 24 hours

Disaster Recovery Plan:

1. Disaster Declaration (IT Manager/Director)
2. Activate DR team
3. Assess damage
4. Restore from backup (priority order)
5. Validate data integrity
6. Switch to DR site (if primary site down)
7. Communication to users
8. Post-incident review

Testing:

- DR drill: Quarterly
- Backup restoration test: Monthly
- Failover test: Annually

10. Deployment Strategy

10.1 Phased Rollout

Phase 1: Pilot (Month 1-3)

Facility: Regional Hospital of Malabo (1 department)

Modules:

- Patient Registration
- Appointment Scheduling
- EMR (basic)
- Billing

Goals:

- Test system in real environment
- Identify issues
- Gather user feedback
- Refine workflows

Success Criteria:

- 90% user adoption
- No critical bugs
- Patient satisfaction maintained

Phase 2: Malabo Expansion (Month 4-6)

Facility: Regional Hospital of Malabo (all departments)

Additional Modules:

- Laboratory

- Radiology
- Pharmacy
- Inpatient Management

Goals:

- Full hospital digitization
- Inter-departmental workflows tested
- Staff fully trained

Phase 3: Multi-Site Rollout (Month 7-12)

Facilities:

- General Hospital of Sampaka
- Regional Hospital of Bata
- Dr. Loeri Comba Polyclinic
- INSESO Bata

Goals:

- Multi-hospital data sharing
- Ministry dashboard operational
- Full system functionality

Phase 4: Advanced Features (Month 13-18)

Modules:

- Advanced Analytics
- Telemedicine (if ready)
- Mobile apps (patient portal)
- Integration with external systems (insurance, etc.)

10.2 Data Migration

Existing Data Migration:

Source: Paper records, Excel sheets, standalone systems

Strategy:

1. Data Assessment
 - Identify what data exists
 - Data quality check
 - Prioritize critical data
2. Data Cleansing
 - Remove duplicates
 - Standardize formats
 - Fix inconsistencies

3. Data Mapping
 - Map old fields to new fields
 - Handle unmapped data
4. Migration Execution
 - Pilot migration (sample data)
 - Validation
 - Full migration
 - Re-validation
5. Cutover
 - Final sync
 - Go-live
 - Old system decommissioned

Timeline: 3 months (parallel to Phase 1)

Data to Migrate:

- Patient demographics (active patients)
- Medical records (last 2 years minimum)
- Staff records
- Inventory (current stock)
- Financial data (current fiscal year)

Historical Data:

- Older records: Scanned and archived (not fully digitized)
- Retrieval on-demand

10.3 Training Program

Training Levels:

Level 1: System Administrators (1 week intensive)

Topics:

- System architecture
- Server management
- Database administration
- Backup/restore procedures
- Security management
- Troubleshooting

Trainees: IT staff from each hospital (10-15 people)

Level 2: Super Users (3 days intensive)

Topics:

- All modules (full functionality)
- Workflow configurations
- Report generation
- User management
- First-level troubleshooting

Trainees: 2-3 per department per hospital (50-75 people)

Role: Train others, provide on-floor support

Level 3: End Users (1 day per role)**Topics:**

- Role-specific training (only relevant modules)
- Common tasks
- Basic troubleshooting

Trainees:

- Doctors: 150 people
- Nurses: 200 people
- Receptionists: 50 people
- Pharmacists: 30 people
- Lab staff: 40 people
- Radiology staff: 25 people
- Billing staff: 25 people

Total: ~500 people

Delivery: On-site, hands-on training

Training Materials:

- User manuals (role-specific)
- Video tutorials
- Quick reference guides (laminated cards)
- FAQs
- Online help system (within application)

Ongoing Training:

- Refresher sessions: Quarterly
- New feature training: As released
- New employee onboarding: Continuous

10.4 Change Management

Communication Plan:

Stakeholders: Ministry officials, hospital directors, doctors, nurses, staff

Messages:

- Why the change? (benefits, government priority)
- What will change? (workflows, processes)
- When? (timeline)
- How? (training, support)

Channels:

- Ministry announcement (official)
- Hospital meetings (all-staff)
- Email newsletters
- Posters in hospitals
- Training sessions

Resistance Management:

- Address concerns (fear of technology, job security)
- Demonstrate benefits (easier work, better patient care)
- Involve champions (early adopters who advocate)
- Provide ample support (hotline, on-site support)

Go-Live Support:

Weeks 1-2 (Critical Period):

- On-site support: Full team at each hospital
- Extended hours: 6 AM - 10 PM
- Hotline: 24/7
- Issue escalation: Immediate

Weeks 3-4:

- On-site support: Reduced team
- Normal hours: 8 AM - 6 PM
- Hotline: 24/7

Month 2+:

- Remote support primary
- On-site visits: As needed
- Hotline: 24/7 (critical issues), business hours (non-critical)

11. Training & Support

11.1 Help Desk / Support System

Tier 1: Help Desk (Local IT)

Coverage: 24/7
Response Time: < 15 minutes
Handles:
- Password resets
- Basic navigation help
- Printer issues
- Network connectivity
- General inquiries

Team Size: 2-3 people per hospital per shift

Tier 2: Application Support (Development Team)

Coverage: Business hours (8 AM - 6 PM) + on-call
Response Time: < 2 hours
Handles:
- Application errors
- Data issues
- Configuration changes
- Report generation
- Integration issues

Team Size: 5-8 people (centralized)

Tier 3: Development Team

Coverage: Business hours + on-call (critical issues)
Response Time: < 4 hours (critical), < 24 hours (normal)
Handles:
- Bug fixes
- System failures
- Database issues
- Complex integrations
- Performance problems

Team Size: 10-15 developers

Issue Tracking:

- Ticketing system (Jira, Zendesk, etc.)
- Priority levels:
 - P1 (Critical): System down, data loss - response < 30 min
 - P2 (High): Major function not working - response < 2 hours
 - P3 (Medium): Minor function not working - response < 8 hours
 - P4 (Low): Enhancement request, cosmetic issue - response < 48 hours
- SLA (Service Level Agreement):

- P1: Resolution < 4 hours
- P2: Resolution < 24 hours
- P3: Resolution < 3 days
- P4: Resolution < 2 weeks

11.2 User Documentation

Manuals:

- System Administrator Guide (500+ pages)
- User Manual per Role (50-100 pages each):
 - Doctor's Guide
 - Nurse's Guide
 - Pharmacist's Guide
 - Lab Technician's Guide
 - Receptionist's Guide
 - Billing Clerk's Guide
- Quick Start Guide (10 pages, laminated)

Video Tutorials:

- 5-minute quick tutorials (50+ videos)
- Screen recordings with voiceover
- Accessible within application (Help menu)

Online Help:

- Context-sensitive help (F1 key)
- Searchable knowledge base
- FAQs

Languages:

- Spanish (primary)
- French (secondary)
- Portuguese (optional)

11.3 Maintenance & Updates

Preventive Maintenance:

- Database optimization: Monthly
- Log file cleanup: Weekly
- Security patches: As released (tested in staging first)
- Certificate renewal: Before expiry

System Updates:

Minor Updates (Bug Fixes, Small Features):
- Frequency: Bi-weekly

- Deployment: Off-hours (2-4 AM)
- Downtime: < 30 minutes
- Notification: 48 hours advance

Major Updates (New Modules, Major Features):

- Frequency: Quarterly
- Deployment: Planned weekend maintenance
- Downtime: 2-4 hours
- Notification: 2 weeks advance
- Training: Provided before deployment

Emergency Patches:

- Critical security vulnerabilities: Immediate
- Critical bugs affecting patient safety: Immediate
- Approval: Expedited

12. Success Metrics (KPIs)

12.1 System Performance Metrics

Technical KPIs:

- System Uptime: Target 99.9%
- Page Load Time: < 2 seconds
- API Response Time: < 500 ms
- Database Query Time: < 100 ms
- Concurrent Users: Support 500+ simultaneous users
- Data Backup Success Rate: 100%
- Failed Login Attempts: < 0.1% (security indicator)

12.2 User Adoption Metrics

Usage KPIs:

- Active Users: Target 90%+ of staff
- Logins per Day: Track growth
- Features Utilized: Target 80%+ of features used
- Mobile App Adoption: Target 50%+ of doctors
- Training Completion Rate: Target 100%

12.3 Operational Efficiency Metrics

Process Improvement KPIs:

Before vs After System:

Patient Registration Time:

- Before: 15-20 minutes
- After: < 5 minutes
- Target: 70% reduction

Appointment Scheduling:

- Before: Phone calls, manual ledger (10 min per appointment)
- After: Online/on-site digital (2 min)
- Target: 80% reduction

Lab Result Turnaround Time:

- Before: 4-8 hours (paper reports)
- After: 1-2 hours (digital)
- Target: 50% reduction

Billing Time:

- Before: 20-30 minutes (manual calculation)
- After: < 5 minutes (auto-calculation)
- Target: 75% reduction

Medication Dispensing Time:

- Before: 15-20 minutes (handwritten prescriptions)
- After: 5-10 minutes (e-prescriptions)
- Target: 50% reduction

Bed Occupancy Rate:

- Before: 60-70% (poor coordination)
- After: 80-85% (optimized)
- Target: 15% increase

12.4 Quality of Care Metrics

Clinical KPIs:

- Medication Error Rate: Target < 1% (reduced via e-prescribing)
- Lab Sample Rejection Rate: Target < 2% (barcode accuracy)
- Hospital-Acquired Infection Rate: Monitor & reduce (better hygiene tracking)
- Readmission Rate (30-day): Monitor & reduce
- Patient Wait Time (OPD): Target < 30 minutes
- Average Length of Stay: Monitor & optimize

12.5 Financial Metrics

Financial KPIs:

- Revenue Growth: Target 20%+ (better billing capture, reduced leakage)
- Collection Rate: Target 95%+ (improved billing efficiency)
- AR Days: Target < 30 days (faster insurance claims)
- Claim Denial Rate: Target < 5% (improved documentation)
- Medication Wastage: Target < 2% (better inventory management)
- Cost per Patient: Monitor & optimize

12.6 Patient Experience Metrics

Patient Satisfaction KPIs:

- Overall Satisfaction Score: Target 85%+
 - Willingness to Recommend: Target 80%+
 - Complaint Rate: Target < 5%
 - Wait Time Satisfaction: Target 75%+
 - Staff Courtesy: Target 90%+
 - Facility Cleanliness: Target 85%+
- Measurement: Patient satisfaction surveys (paper/digital)

12.7 Ministry/Strategic Metrics

National Health KPIs:

- Total Patients Treated: Track growth
- Geographic Coverage: Ensure equitable access
- Disease Surveillance: Early outbreak detection
- Resource Utilization: Optimize allocation
- Healthcare Access: Reduce disparities
- Data Completeness: Target 95%+ (for policy decisions)

13. Budget Estimate (High-Level)

Note: Actual costs will be determined in detailed project proposal

13.1 Software Development

- Phase 1: Core Modules Development (6 months)
- Development Team (10-15 developers)
 - UI/UX Design
 - Backend Development
 - Mobile App Development
 - Testing & QA
- Estimated: \$300,000 - \$500,000

Phase 2: Advanced Modules (6 months)

- Additional features
- Integrations
- Reporting & Analytics

Estimated: \$200,000 - \$350,000

Phase 3: Refinements & Optimization (6 months)

- Performance optimization
- Security hardening
- User feedback implementation

Estimated: \$150,000 - \$250,000

Total Software Development: \$650,000 - \$1,100,000

13.2 Infrastructure

Cloud Infrastructure (Annual):

- Servers (compute, storage)
- Bandwidth
- Backup storage

Estimated: \$100,000 - \$200,000/year

OR

On-Premise Infrastructure (One-time):

- Servers (application, database, storage)
- Networking equipment
- UPS & generators
- Server room setup

Estimated: \$250,000 - \$400,000

Client Devices (One-time):

- Desktops (200 units): \$100,000
- Tablets (50 units): \$25,000
- Printers (barcode, receipt, paper): \$50,000
- Scanners, accessories: \$25,000

Total: \$200,000

13.3 Training & Change Management

Training Program:

- Super user training (75 people x 3 days)
- End user training (500 people x 1 day)
- Training materials development
- Trainers' fees
- Venue, logistics

Estimated: \$100,000 - \$150,000

```
Change Management:
- Communication campaigns
- On-site support during go-live
- Champions program
Estimated: $50,000 - $80,000
```

13.4 Annual Operating Costs

```
Year 1 (Implementation Year):
- Development: $650,000 - $1,100,000
- Infrastructure: $250,000 - $400,000 (on-premise) OR $100,000-$200,000 (cloud)
- Client devices: $200,000
- Training: $100,000 - $150,000
- Change management: $50,000 - $80,000
TOTAL YEAR 1: $1,250,000 - $1,930,000

Year 2+ (Annual Operating):
- Maintenance & support: $150,000 - $250,000
- Infrastructure (cloud): $100,000 - $200,000 OR (on-premise): $50,000
- Staff training (new hires, refreshers): $30,000 - $50,000
- Upgrades & enhancements: $100,000 - $200,000
TOTAL YEAR 2+: $380,000 - $700,000/year
```

13.5 Funding Options

- Government budget allocation
- World Bank / IMF loans
- WHO / UNICEF grants
- Public-Private Partnership (PPP)
- Phased investment (start with 1-2 hospitals, expand gradually)

14. Risk Management

14.1 Identified Risks

Technical Risks:

```
Risk: Internet connectivity issues in remote areas
Mitigation: Offline mode with auto-sync
```

```
Risk: Data loss
Mitigation: Redundant backups, disaster recovery plan
```

```
Risk: System performance issues (too slow)
Mitigation: Performance testing, scalability architecture
```

Risk: Security breaches

Mitigation: Security audits, encryption, access controls

Operational Risks:

Risk: User resistance to change

Mitigation: Change management, training, super users

Risk: Insufficient training

Mitigation: Comprehensive training program, ongoing support

Risk: Staff turnover (trained staff leave)

Mitigation: Documentation, refresher training, video tutorials

Risk: Power outages

Mitigation: UPS, generators, battery-powered devices

Project Risks:

Risk: Scope creep (too many feature requests)

Mitigation: Clear requirements, change control process

Risk: Budget overruns

Mitigation: Phased approach, prioritization, contingency budget

Risk: Timeline delays

Mitigation: Agile methodology, buffer time, risk monitoring

Risk: Vendor dependency

Mitigation: Open standards, source code ownership, documentation

External Risks:

Risk: Regulatory changes

Mitigation: Flexible architecture, compliance monitoring

Risk: Economic factors (currency fluctuation)

Mitigation: Fixed-price contracts, local vendors where possible

Risk: Political factors (change in government priorities)

Mitigation: Legal agreements, contracts, ministerial approval

15. Project Timeline (Gantt Chart Summary)

YEAR 1

Month 1-3: Requirements & Design

- | Stakeholder workshops
- | Detailed requirements gathering
- | System design & architecture
- | Database design
- | UI/UX mockups

Month 4-9: Development Phase 1 (Core Modules)

- | Patient Management
- | Appointment Scheduling
- | EMR
- | Billing
- | User Management
- | Basic Reporting

Month 10-12: Development Phase 2 (Clinical Modules)

- | Pharmacy Management
- | Laboratory Information System
- | Radiology & PACS
- | Inpatient Management
- | Emergency Department

Month 10-12 (Parallel): Infrastructure & Pilot

- | Server setup
- | Network configuration
- | Pilot at Malabo Regional (1 dept)
- | Super user training
- | Feedback collection

YEAR 2

Month 13-15: Development Phase 3 (Advanced Modules)

- | Operation Theatre Management
- | Maternity & Obstetrics
- | Advanced Analytics
- | Ministry Dashboard
- | Mobile Apps

Month 13-18: Rollout

- | Month 13-14: Malabo Regional (full hospital)
- | Month 15: Sampaka Hospital
- | Month 16: Bata Regional Hospital
- | Month 17: Dr. Loeri Polyclinic
- | Month 18: INSESO Bata
- | Ongoing training & support

Month 19-24: Stabilization & Optimization

- | Bug fixes
- | Performance optimization
- | User feedback implementation
- | Integration with external systems
- | Telemedicine (if ready)

16. Appendices

Appendix A: Glossary

ADR: Adverse Drug Reaction
ANC: Antenatal Care
API: Application Programming Interface
ARPOB: Average Revenue Per Occupied Bed
ASA: American Society of Anesthesiologists (classification)
CBC: Complete Blood Count
DICOM: Digital Imaging and Communications in Medicine
EDD: Expected Date of Delivery
EHR: Electronic Health Record
EMR: Electronic Medical Record
ESI: Emergency Severity Index
FHIR: Fast Healthcare Interoperability Resources
HIS: Hospital Information System
HL7: Health Level 7 (interoperability standard)
ICD-10: International Classification of Diseases, 10th Revision
ICU: Intensive Care Unit
LIS: Laboratory Information System
LMP: Last Menstrual Period
LOINC: Logical Observation Identifiers Names and Codes
MAR: Medication Administration Record
MRI: Magnetic Resonance Imaging
NHMS: National Healthcare Management System
OPD: Outpatient Department
OT: Operation Theatre
PACS: Picture Archiving and Communication System
PMS: Patient Management System
RIS: Radiology Information System
RPO: Recovery Point Objective
RTO: Recovery Time Objective
SLA: Service Level Agreement
SOAP: Subjective, Objective, Assessment, Plan (note format)
TPR: Temperature, Pulse, Respiration
VPN: Virtual Private Network
WHO: World Health Organization

Appendix B: Sample Forms

(This section would include sample forms like):

- Patient Registration Form
- Consent Form
- Prescription Template
- Lab Request Form
- Discharge Summary Template

- etc.

(Not included here for brevity, but would be in final document)

Appendix C: Workflow Diagrams

(This section would include detailed workflow diagrams like):

- Patient Registration Workflow
- Lab Order to Result Workflow
- Admission to Discharge Workflow
- Emergency Department Workflow
- Pharmacy Dispensing Workflow
- etc.

(Not included here for brevity, but would be in final document)

Appendix D: Screen Mockups

(This section would include UI mockups like):

- Dashboard (Doctor, Nurse, Admin)
- Patient Registration Screen
- EMR Screen
- Appointment Calendar
- Lab Result Entry Screen
- Pharmacy Dispensing Screen
- etc.

(Not included here for brevity, but would be in final document)

Appendix E: Technical Architecture Diagrams

(This section would include):

- System Architecture Diagram
- Network Topology
- Database Schema (ER Diagram)
- Integration Architecture
- Security Architecture
- Deployment Architecture
- etc.

(Not included here for brevity, but would be in final document)

17. Conclusion

The National Healthcare Management System represents a transformative initiative for the healthcare sector of Equatorial Guinea. This comprehensive system will:

1. **Improve Patient Care:** Through integrated medical records, clinical decision support, and coordinated care across facilities
2. **Increase Efficiency:** By automating manual processes, reducing paperwork, and streamlining workflows
3. **Enable Data-Driven Decisions:** By providing real-time analytics and reporting for evidence-based policy making
4. **Ensure Financial Sustainability:** By improving revenue capture, reducing wastage, and optimizing resource allocation
5. **Support National Development:** As a critical component of the National Health Development Plan, serving the entire population

Next Steps:

1. Ministry approval of this specification
2. Detailed project proposal with cost breakdown
3. Vendor selection / contract finalization
4. Project kickoff
5. Implementation per timeline

Success Factors:

- Strong ministerial support and oversight
- Adequate funding and resources
- Comprehensive training and change management
- Robust technical infrastructure
- Ongoing maintenance and support
- User engagement and feedback

This project will position Equatorial Guinea as a leader in digital health in the region and ensure high-quality medical care for all citizens.

Document Prepared By: [Your Company Name] **Date:** 22 May 2026 **Version:** 1.0 - Draft for Ministry Review

Status: Awaiting Ministry Approval

For Official Use:

Ministry of Health Approval:

Minister of Health Date: _____

Secretary General Date: _____

END OF DOCUMENT